

How to construct even faster and indifferentiable hash functions from random permutations

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Abstract. Designing cryptographic hash functions that simultaneously achieve high throughput and strong provable security remains a fundamental challenge in symmetric cryptography. Traditional constructions, such as the Merkle-Damgård (MD) and SPONGE paradigms, inherently suffer from structural limitations: they either expose internal chaining values to length-extension attacks or impose a rigid trade-off between security (capacity) and efficiency (rate), bounding their architectural potential.

In this paper, we introduce the **Compress-then-Randomize** paradigm, a modular design principle that structurally decouples a hash function into two independent components with distinct security objectives: (1) a *variable-input-length* (VIL) compression component optimized for high-speed message absorption, requiring only collision resistance and multiple-preimage resistance; and (2) a *fixed-input-length* (FIL) finalization component utilizing independent random permutations to ensure indistinguishability from a random oracle. This separation enables the VIL component to maximize processing throughput (approaching the full primitive width) while the FIL component provides robust randomness extraction, effectively mitigating length-extension attacks and achieving tight security bounds.

Leveraging this paradigm, we propose the **Rocket hash family**, comprising two instantiations: *Rocket-JH* (based on the JH iteration structure with wide-pipe design) and *Rocket-DoubleCBC* (utilizing dual parallel Cipher-Block-Chaining lanes). Both constructions employ pairwise distinct round permutations and achieve superior message processing rates compared to conventional SPONGE-based designs, with Rocket-2 delivering more than $2\times$ the throughput of SHA3-512 for large messages.

For practical instantiation without requiring multiple independent cryptographic primitives, we present **CTR-Perm**, a novel domain-separation technique that derives 2^w effectively independent round functions from a single large permutation using a counter-based input diversification. While backward queries introduce a heuristic assumption regarding preimage multiplicity (bounded by a negligible failure probability $\leq 2^{-526}$ for counter size $w \geq 64$ and hash length 512 bits), we prove the construction remains sound for practical message lengths up to 2^{w-8} blocks.

Finally, to formalize the security-efficiency trade-off, we propose **Hash Effectiveness** (H.E.), a scale-invariant heuristic metric defined as the product of normalized security level and normalized processing rate. We demonstrate that conventional Merkle-Damgård and SPONGE constructions exhibit H.E. values fundamentally bounded by $1/8$, whereas the Rocket constructions achieve H.E. values approaching $1/4$ (specifically, 0.227 for Rocket-2 with Keccak- $p[1600]$), thereby establishing a new Pareto frontier in hash function design.

Keywords: Hash Function · Permutation-Based Cryptography · Indistinguishability · Provable Security · Domain Separation

1 Introduction

1.1 Background

Hash Functions are fundamental cryptographic primitives with broad applications in both practical and theoretical domains. Designing cryptographically secure hash functions presents a fundamental challenge rooted in mathematics. This challenge arises from the inherent contradiction between security requirements and the structural constraints of hash functions: they must map an infinite domain (arbitrary-length messages) onto a finite range (fixed-length digests) while maintaining robust security properties such as collision resistance and preimage resistance. Consequently, practical hash functions can only achieve *computational security*. This means that while insecure events (e.g., collisions or preimages) mathematically exist, no polynomial-time adversary with limited computational power can feasibly find them under standard cryptographic assumptions. Thus, such hash functions are deemed secure for practical applications. With respect to efficiency, the primary challenge lies in the speed at which a hash function compresses variable-length messages into fixed-length digests. This process relies on an iterative structure where the input is divided into fixed-size blocks, each processed sequentially by a compression function.

Merkle-Damgård Construction provides an intuitive method for iterating a compression function f to process arbitrary-length messages [16,34]. Given message blocks $m_1, m_2, \dots, m_l$, the hash values are computed iteratively as:

$$h_i = f(h_{i-1}, m_i) \quad \text{for } i = 1, 2, \dots, l, \quad \text{with } h_0 = IV.$$

A key security guarantee of this design is that if the underlying compression function is collision-resistant, the resulting hash function also maintains collision resistance. Consequently, the MD structure underpins widely-used hash functions such as MD4, MD5, SHA-1, and SHA-2.

However, the MD construction exhibits several structural vulnerabilities. It is susceptible to *length-extension attacks*, wherein an adversary who obtains $\text{MD}(M)$ can compute $\text{MD}(M||X)$ for some suffix X without knowing M . Countermeasures include MD strengthening (appending the message length) [16,34], the wide-pipe design [31], and using independent finalization functions as in NMAC and EMAC [8,37]. Furthermore, the MD construction allows for efficient *multi-collision attacks*; Joux showed that finding 2^t collisions requires only $t \cdot 2^{n/2}$ work, significantly less than the ideal 2^n complexity [23]. Kelsey and Schneier further demonstrated that length padding alone is insufficient against advanced attacks using expandable messages [26].

More fundamentally, the MD construction is not indifferentiable from a random oracle [15,32], limiting its theoretical soundness in random-oracle-based proofs. Proposed fixes include output truncation, suffix encoding [15], or incorporating an independent permutation during finalization [21]. Dodis et al. introduced the concept of *preimage awareness* (PrA), showing that a PrA-compliant hash function composed with a fixed-input-length random oracle can achieve variable-input-length random oracle properties in the indifferentiability framework [19,32]. Nevertheless, many adjustments either compromise cryptographic strength or introduce practical inefficiencies.

SPONGE Construction achieves inherent resistance against length-extension attacks by design, as it refrains from exposing internal chaining values directly. Instead, it bifurcates the permutation state into two distinct segments: an r -bit *rate* segment utilized for absorbing message blocks and subsequently squeezing hash output, and a c -bit *capacity* segment functioning as a cryptographically protected internal state that remains entirely unexposed throughout the

computation [11]. The total state size $b = r + c$ is selected to be sufficiently large—exemplified by SHA-3’s adoption of $b = 1600$ with varying rate-capacity allocations [2]—ensuring that r maintains practical processing efficiency while c provides the requisite security margin against generic attacks.

Notably, the SPONGE construction mitigates multi-collision attacks by fundamentally elevating the complexity of finding initial collisions, as the security level is intrinsically tethered to the capacity c . Specifically, the construction has been formally proven to be indistinguishable from a random oracle up to the birthday bound of approximately $2^{c/2}$ queries [11]. This rigorous security guarantee provides the theoretical foundation for deploying SPONGE-based hash functions in cryptographic protocols whose security analyses presuppose the availability of a random oracle [32].

Compared to the Merkle-Damgård paradigm, the SPONGE construction offers distinct architectural advantages, most prominently native support for flexible output lengths (extendable-output functions) and inherent multi-functionality. These characteristics render it suitable for diverse applications beyond conventional hashing, including message authentication codes, authenticated encryption schemes, stream ciphers, and pseudorandom number generators. This versatility has catalyzed extensive research into generalizations and extensions, such as full-state keyed SPONGE, duplex constructions, and beyond-birthday-bound variants [5,25,33,28,29], continually expanding the applicability and fortifying the security foundations of permutation-based primitives.

The Rate-Capacity Trade-off. A fundamental limitation inherent to the SPONGE construction stems from the fixed size $b = r + c$ of its underlying permutation. This constraint imposes an inescapable trade-off between security and efficiency: enhancing the capacity c to improve the provable security bound (approximately $2^{c/2}$) necessarily diminishes the rate $r = b - c$, thereby constraining the message absorption throughput. While this flexibility allows parameterization for diverse security requirements, it fundamentally caps the achievable performance. Formally, the product of the normalized security level $c/(2b)$ and the normalized rate $(b - c)/b$ is bounded by:

$$\frac{c}{2b} \times \frac{b - c}{b} = \frac{c(b - c)}{2b^2} \leq \frac{1}{8},$$

where the maximum is attained when $c = b/2$. This inequality establishes a theoretical performance ceiling of $1/8$ for the Hash Effectiveness metric, which we analyze comprehensively in Section 6.

Attempts to transcend this limitation include the Double SPONGE variant [29], which executes two independent SPONGE instances in parallel to achieve beyond-birthday-bound security of approximately $2^{2c/3}$ against certain classes of attacks. However, this security enhancement extracts a substantial cost in efficiency: the effective rate is necessarily halved to $r/2$, as each message block must be processed twice (once per instance), thereby shifting the trade-off toward higher security at the expense of computational throughput.

Tree-based Constructions leverage parallel processing to achieve high throughput on well-equipped platforms. However, their practical adoption remains limited due to structural complexity and inefficiencies in handling non-uniform message lengths [16,34,39].

1.2 Our Contributions

In this paper, we continue the investigation into permutation-based hash constructions, focusing on overcoming the fundamental rate-capacity trade-off inherent in existing paradigms. Our work

is motivated by the key observation that *a fixed-input-length (FIL) random oracle consistently produces uniformly random outputs provided its inputs are pairwise distinct*. Building upon this insight, we posit that a variable-input-length (VIL) hash function with ideal output behavior can be constructed by architecturally decoupling the computation into two phases: a high-speed iterative compression phase that ensures structural integrity (collision and preimage resistance), followed by a cryptographically strong finalization phase that enforces random-oracle-like unpredictability. Specifically, our contributions are fourfold:

1. **The Compress-then-Randomize Paradigm.** We introduce the *Compress-then-Randomize* paradigm, a modular design methodology that structurally separates hash computation into independent VIL and FIL components to achieve provable indistinguishability from a random oracle. The core insight is that an independent finalization phase can effectively neutralize structural vulnerabilities such as length-extension attacks while maintaining compositional security. We instantiate this paradigm through a concrete framework named **Decohash**, and rigorously prove that it can upgrade any VIL hash function satisfying collision resistance and multiple-preimage resistance into a full indistinguishable construction. This modularity enables independent optimization of the compression phase for throughput without compromising final output randomness.
2. **The Rocket Hash Family.** Guided by the Decohash framework, we propose the *Rocket* hash family, comprising two primary instantiations: *Rocket-JH* (enhancing the JH iteration structure with wide-pipe design) and *Rocket-DoubleCBC* (utilizing dual independent Cipher-Block-Chaining lanes operating in parallel). Both designs strictly enforce the separation between the iterative message absorption phase and the finalization phase, employing *pairwise distinct* random permutations for each round. This architectural choice facilitates tighter indistinguishability proofs by eliminating problematic query cycles between the VIL and FIL components. The use of independent per-round permutations represents a departure from conventional designs, though it introduces non-trivial implementation challenges regarding permutation diversity.
3. **The CTR-Perm Construction.** To address the practical challenge of instantiating multiple independent permutations without prohibitive hardware costs, we present **CTR-Perm**, a novel domain-separation technique that derives 2^w effectively independent random functions from a single large-scale permutation via counter-based input diversification. While CTR-Perm does not yield fully independent random permutations under adversarial backward queries—introducing a heuristic assumption regarding preimage multiplicity bounded by a negligible failure probability—we rigorously prove its sufficiency for securely instantiating the Rocket hash constructions. Specifically, we demonstrate that for counter size $w \geq 64$, the construction remains sound under practical message-length constraints (up to 2^{w-8} blocks), with the backward-query amplification factor bounded by a small constant ($u \leq 128$) with overwhelming probability. This technique may additionally inspire new primitives requiring efficient derivation of multiple function instances from a single cryptographic component.
4. **Definition and Analysis of Hash Effectiveness.** To formalize the evaluation of security-efficiency trade-offs in hash constructions, we introduce **Hash Effectiveness (H.E.)**, a scale-invariant heuristic metric defined as the product of normalized security level and normalized processing rate:

$$\text{H.E.} = \underbrace{\frac{\text{Provable Security Level (bits)}}{b}}_{\text{Security Ratio}} \times \underbrace{\frac{\text{Processing Rate (bits per invocation)}}{b}}_{\text{Efficiency Ratio}},$$

where b denotes the primitive size. Comparative analysis reveals that conventional Merkle-Damgård and SPONGE constructions exhibit an apparent theoretical ceiling of $\text{H.E.} \leq 1/8$, arising from the inherent rate-capacity tension. We affirmatively demonstrate that the Rocket hash family transcends this limitation: Rocket-JH achieves H.E. approaching $3/16$, while Rocket-DoubleCBC attains values approaching $1/4$, thereby confirming that structural decoupling fundamentally expands the achievable Pareto frontier of the security-efficiency trade-off.

1.3 Related Works

The HAIFA (HAsH Iterative FrAmework) framework enhances MD by integrating salt and message length information directly into the compression function rounds. This design mitigates specific attacks, while offering built-in support for randomized hashing and variable output sizes [13]. Hash combiners aim to bolster security by combining multiple hash functions. Unfortunately, many combiners exhibit critical weaknesses [7]. Further contributions include refined security notions for hash functions [40,9,4], innovative designs for compression functions [38,14], and extensive cryptanalytic studies that reveal vulnerabilities in iterative structures [23,24,3,6].

1.4 Organization of This Paper

The remainder of this paper is structured as follows. Section 2 introduces the necessary notation and preliminary concepts. Section 3 presents the general Compress-then-Randomize paradigm and proves the indifferenciability security of a specific instantiation named Decohash. Section 4 details the Rocket hash family, comprising the Rocket-JH and Rocket-DoubleCBC designs, and provides their corresponding security bounds. Section 5 describes the CTR-Perm technique and presents security results for Rocket hash members built upon it. Section 6 formally defines hash effectiveness and conducts a comparative analysis of permutation-based hash constructions and the ISO-standardized hash functions using this metric. Finally, Section 7 concludes the paper.

2 Notations and Security Definitions

This section establishes the fundamental notation and security definitions used throughout our analysis. We adopt standard cryptographic conventions and refer to [40,32,15] for comprehensive treatments.

2.1 Notations

We denote by $\{0,1\}^*$ the set of all finite-length binary strings and by $\{0,1\}^n$ the set of n -bit strings. For strings x and y , concatenation is written as $x \parallel y$, and $|x|$ denotes the bit-length of x . The bitwise XOR operation is denoted by $x \oplus y$ (defined only when $|x| = |y|$). For a positive integer $a \leq |x|$, $\text{LMB}(a, x)$ and $\text{RMB}(a, x)$ extract the leftmost and rightmost a bits of x , respectively.

A *random permutation* P over $\{0,1\}^n$ is a permutation selected uniformly at random from the set of all $(2^n)!$ permutations on n -bit strings; its inverse is denoted P^{-1} . A *random function* RF from $\{0,1\}^*$ to $\{0,1\}^n$ is a function where each distinct input x is assigned an output sampled independently and uniformly from $\{0,1\}^n$.

A *hash function* is a deterministic algorithm $H : \{0,1\}^* \rightarrow \{0,1\}^h$ that maps arbitrary-length messages to fixed-length digests of h bits. The security notions for hash functions originate

from the seminal works of Merkle and Damgård [16,34]. This paper focuses on the following fundamental properties: collision resistance, preimage resistance, multiple-preimage resistance, and indistinguishability from a random oracle.

2.2 Security Definitions

The security advantages defined below are parameterized by q , the number of the adversary's queries to the underlying primitives (compression functions or public permutations) of hash function H . The maximum advantage is taken over all adversaries $\mathcal{A}$ restricted to q .

Collision Resistance. A hash function H is *collision-resistant* if it is computationally infeasible to find two distinct messages that hash to the same value. Formally, the advantage of an adversary $\mathcal{A}$ is:

$$\text{Adv}_H^{\text{coll}}(q) = \max_{\mathcal{A}} \Pr \left[\mathcal{A}^{H(\cdot)} \rightarrow (M, M') : M \neq M' \wedge H(M) = H(M') \right].$$

Preimage Resistance. A hash function H is *preimage-resistant* if, given a uniformly random hash value y , it is computationally infeasible to find any message that hashes to y . Formally:

$$\text{Adv}_H^{\text{pre}}(q) = \max_{\mathcal{A}} \Pr \left[y \xleftarrow{\$} \{0, 1\}^h; \mathcal{A}^{H(\cdot)}(y) \rightarrow M : H(M) = y \right].$$

Multiple-Preimage Resistance. A hash function H is *multiple-preimage-resistant* if, given p independent target hash values, it is computationally infeasible to find a message that hashes to any of them. Formally:

$$\text{Adv}_H^{\text{m-pre}}(p, q) = \max_{\mathcal{A}} \Pr \left[y_1, \dots, y_p \xleftarrow{\$} \{0, 1\}^h; \mathcal{A}^{H(\cdot)}(y_1, \dots, y_p) \rightarrow M : \exists i \in [1, p], H(M) = y_i \right].$$

Second-Preimage Resistance. A hash function H is *second-preimage-resistant* if, given a uniformly random challenge message M , it is computationally infeasible to find a different message M' with the same hash value. Formally:

$$\text{Adv}_H^{2\text{nd-pre}}(q) = \max_{\mathcal{A}} \Pr \left[M \xleftarrow{\$} \{0, 1\}^*; \mathcal{A}^{H(\cdot)}(M) \rightarrow M' : M \neq M' \wedge H(M) = H(M') \right].$$

Indifferentiability. Indifferentiability [32,15] provides a stronger security guarantee: a hash construction H^f built from an ideal primitive f should be indistinguishable from a random oracle $\mathcal{R}$, even when the adversary has oracle access to the underlying primitive f . This notion is formalized through the two-world paradigm:

- *Real World:* The adversary interacts with both the hash construction H^f (implemented using primitive f) and the primitive f itself.
- *Ideal World:* The adversary interacts with a random oracle $\mathcal{R}$ and a simulator $\mathcal{S}$ that simulates the primitive f while maintaining consistency with $\mathcal{R}$.

The hash construction H^f is *indifferentiable* from a random oracle if there exists an efficient simulator $\mathcal{S}$ such that no computationally-bounded distinguisher $\mathcal{D}$ can non-negligibly differentiate between the two worlds. Formally, the indifferentiability advantage is:

$$\text{Adv}_{(H^f, f)}^{\text{indiff}}(q_{\mathcal{S}}, q_{\mathcal{R}}) = \max_{\mathcal{D}} \left| \Pr[\mathcal{D}^{H^f, f} = 1] - \Pr[\mathcal{D}^{\mathcal{R}, \mathcal{S}^{\mathcal{R}}} = 1] \right|$$

where the distinguisher makes at most $q_{\mathcal{S}}$ queries to $\mathcal{S}$ and at most $q_{\mathcal{R}}$ queries to $\mathcal{R}$. This notion ensures that the hash construction can securely replace a random oracle in any cryptographic scheme.

3 A New Framework

3.1 The Compress-then-Randomize Paradigm

Motivation and Core Insight. Traditional hash constructions, such as Merkle-Damgård and the SPONGE construction, typically employ identical round functions across both iterative and finalization phases. Proving indistinguishability for such homogeneous designs requires establishing that the internal state remains pseudorandom in every round—a property that becomes difficult to guarantee once the random chain is broken by collisions or other structural artifacts. While the SPONGE construction manages this through its rate-capacity separation, other designs face fundamental limitations. Notably, the JH hash function, despite using an n -bit permutation, has only been proven to achieve $n/4$ -bit indistinguishability security [35], significantly below the ideal $n/2$ bound.

Structural Decoupling. The Compress-then-Randomize paradigm resolves this tension through architectural decoupling: we explicitly separate the hash function into two independent components with distinct security objectives and optimization targets:

1. **Variable-Input-Length (VIL) Component:** Responsible for compressing arbitrary-length messages into a fixed-size state. This component requires only *collision resistance* and *preimage resistance*—weaker but more efficiently achievable properties than full randomness. By relaxing the security requirement, we enable aggressive optimizations such as higher-rate message absorption, parallel processing lanes, or simplified round functions.
2. **Fixed-Input-Length (FIL) Component:** A dedicated finalization phase that transforms the VIL output into the final hash digest. This component must provide strong randomness guarantees (indistinguishability from a random oracle) but operates on short, fixed-length inputs, allowing the use of computationally intensive yet provably secure randomized functions.

The critical insight is that *independence* between these components—achieved through distinct primitives and isolated state spaces—enables modular security analysis. The VIL component’s weaknesses cannot propagate to compromise the FIL component’s output randomness, while the FIL component’s strength compensates for the VIL component’s relaxed guarantees.

Primitive Selection: RFs vs. RPs. For the FIL component, we consider two classes of randomized functions:

- **Public keyless random functions (RFs):** Offer ideal security properties, exhibiting perfect randomness and independence across all distinct inputs. However, constructing keyless RFs that are both efficiently computable and cryptographically secure is notoriously difficult: deterministic designs inevitably introduce structural artifacts or algebraic regularities that enable distinguishing attacks.
- **Public keyless random permutations (RPs):** Provide mature security analysis, efficient implementations, and widespread adoption in standardized designs including KECCAK, ASCON, and SHA-3 [11,2,17,1]. Their bidirectional structure (permitting both forward and inverse queries) necessitates careful handling but offers superior performance characteristics.

We adopt the *Compress-then-RP* pattern, leveraging the efficiency and analytical maturity of permutation-based cryptography.

Mitigating Inversion Attacks. The use of RPs introduces a specific vulnerability: an adversary with access to the inverse permutation P^{-1} may attempt to invert the finalization function and recover the VIL component’s output state. Successful inversion would enable distinguishing attacks by revealing structural properties of the iterative compression phase. We address this through two established countermeasures:

1. **Output Truncation:** The FIL component outputs only a prefix of the permutation output, discarding sufficient entropy to render inversion computationally infeasible. Formally, for permutation $P : \{0,1\}^n \rightarrow \{0,1\}^n$ and truncation to $h \leq n/2$ bits, the probability of correctly guessing the truncated portion is bounded by $2^{-(n-h)}$.
2. **Feedforward Operation:** The permutation output is combined with its input via XOR: $y = P(x) \oplus x$. This structure ensures that inversion requires solving $y = P(x) \oplus x$ for unknown x —a problem computationally hard for random permutations without knowledge of their internal structure. The feedforward mechanism effectively destroys the algebraic regularity that would otherwise enable efficient inverse computation.

These countermeasures transform the RP into a one-way function from the adversary’s perspective, restoring the security properties necessary for indistinguishability while preserving implementation efficiency.

Theoretical Implications. This paradigm shift enables constructions that surpass the security-efficiency trade-offs inherent in homogeneous designs. By allowing the VIL component to optimize for throughput (maximizing the rate of message absorption) while the FIL component optimizes for security (maximizing output randomness), we achieve a *Pareto improvement* [36,10] over traditional constructions. The formal analysis in subsequent sections demonstrates that this decoupling yields tight, tractable security bounds that scale favorably with primitive size, ultimately enabling the construction of hash functions with superior *Hash Effectiveness*—a metric quantifying the security-efficiency frontier.

3.2 The Decohash Framework

Let $P_0, P_1, \dots, P_z$ be independent random permutations over $\{0,1\}^n$, and let $H[P_0] : \{0,1\}^* \rightarrow \{0,1\}^n$ be any hash construction based on P_0 . For any message $M \in \{0,1\}^*$, its output is denoted as $x_{0,1} \parallel x_{0,2} = H[P_0](M)$. The finalization function $F[P_1, P_2, \dots, P_z] : \{0,1\}^n \rightarrow \{0,1\}^{zh}$ is defined as

$$F(x_{0,1} \parallel x_{0,2}) = x_{1,1} \parallel x_{2,1} \parallel \dots \parallel x_{z,1},$$

where $x_{i,1} \parallel x_{i,2} = P_i(x_{i-1,1} \parallel x_{i-1,2})$ for $i = 1, 2, \dots, z$, with $|x_{i,1}| = h \leq n/2$ and $|x_{i,2}| = n - h$. The composite Decohash (a decoupling hash framework) is $F \circ H : \{0,1\}^* \rightarrow \{0,1\}^{zh}$, as illustrated in Fig. 1.

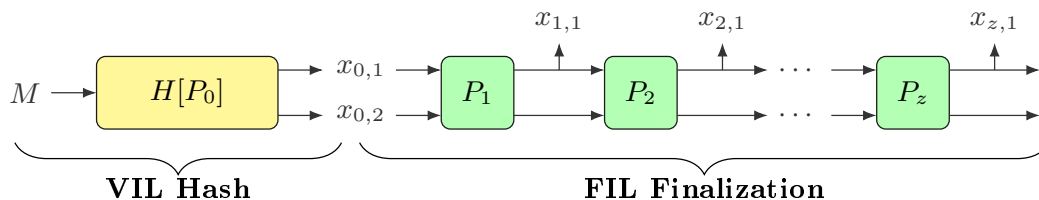


Fig. 1. Decohash Framework: VIL-Hash and FIL-Finalization Architecture

Public Permutations and Extractability. A distinctive feature of the Decohash framework is that all underlying primitives $P_0, P_1, \dots, P_z$ are *publicly accessible permutations*, meaning both the adversary $\mathcal{A}$ and the simulators $\mathcal{S}_i$ may issue forward and backward queries to any P_i and observe the complete query history. Consequently, each simulator $\mathcal{S}_i$ maintains a full transcript of all interactions with its assigned permutation.

This public-query model grants a critical capability absent in traditional indistinguishability scenarios (e.g., the $f(h(\cdot))$ composition of Coron et al. [15]): *extractability*. Because the specifications of H and F define deterministic computation paths over the permutations, any query to the high-level constructions $H[P_0](M)$ or $F[P_1, \dots, P_z](x)$ can be immediately *recognized* and *linked back* to its origin by examining the query history of the underlying P_i 's. Specifically, when $\mathcal{S}_i$ observes a query x to P_1 (or P_1^{-1}), it can determine whether x was previously output by $H[P_0]$ on some message M by scanning the transcript of P_0 , thereby extracting the internal state chain and linking it to the random oracle response $\mathcal{R}(M)$.

In contrast, in the $f(h(\cdot))$ scenario (in Section 3 of [15]), $h(\cdot)$ is a public hash function whose internal computation is opaque to the simulator $\mathcal{S}_f$ simulating f . When $\mathcal{S}_f$ receives a query x to f , it cannot determine the preimage M such that $x = h(M)$, as it lacks visibility into h 's query history. The Decohash framework eliminates this “query unlinkability” problem by exposing the entire primitive interface, allowing the simulator to reconstruct the complete evaluation trace from message M to final output through the recorded permutation queries.

Indifferentiability Proof and Simulator Construction. To prove that Decohash is indifferentiable from a random oracle, we construct a simulator $\mathcal{S} = \{\mathcal{S}_0, \mathcal{S}_1, \dots, \mathcal{S}_z\}$, where each $\mathcal{S}_i$ simulates the permutation P_i for $i \in [0, z]$. A distinguisher $\mathcal{D}$ attempts to tell Decohash apart from a random oracle $\mathcal{R} : \{0, 1\}^* \rightarrow \{0, 1\}^{zh}$. In the real world, $\mathcal{D}$ has oracle access to P_i (and their inverses) and $F \circ H$; in the ideal world, P_i is replaced by $\mathcal{S}_i$ and $F \circ H$ by $\mathcal{R}$.

The simulator $\mathcal{S}$ operates as follows:

- **Initialization:** $\mathcal{S}$ maintains sets to record input-output pairs for each P_i and $H[P_0]$, and internal states for F . These sets are initially empty. Responses to repeated queries are consistent.
- **Random Oracle Simulation:** $\mathcal{S}$ uses a private random oracle (via individual secret keys and counters input to $\mathcal{R}$) to generate random responses, truncating outputs if necessary.
- **Simulating $\mathcal{S}_0$ (for P_0):** $\mathcal{S}_0$ simulates P_0 as a random permutation. After each response, $\mathcal{S}$ checks for collisions or preimage events in H . If a preimage event occurs (i.e., an output matches an entry in a special set BadX , which records queries to P_1 or P_1^{-1} not previously output by H), $\mathcal{S}$ aborts.
- **Simulating $\mathcal{S}_1$ (for P_1 and P_1^{-1}):**
 - On a forward query x_0 to P_1 : If x_0 is a previous output of H for some message M , $\mathcal{S}_1$ retrieves $r_1 \dots r_z = \mathcal{R}(M)$, constructs an internal state chain $[M, x_0, (x_{1,1}, x_{1,2}), \dots, (x_{z,1}, x_{z,2})]$ with $x_{i,1} = r_i$ and $x_{i,2}$ undefined. Then, it randomly selects $x_{1,2}$ consistent with P_1 , and returns $(x_{1,1}, x_{1,2})$. Otherwise, it returns a random consistent output and adds x_0 to BadX .
 - On a backward query $(x_{1,1}, x_{1,2})$ to P_1^{-1} : $\mathcal{S}_1$ returns a random consistent x_0 , adds x_0 to BadX , and aborts if x_0 matches a previous output of H (event $[\text{mitm}]$).
- **Simulating $\mathcal{S}_i$ (for P_i and P_i^{-1} , $i \geq 2$):**
 - On a forward query $(x_{i-1,1}, x_{i-1,2})$: If it links to a valid internal state chain for some M , $\mathcal{S}_i$ sets $x_{i,1} = r_i$ (from $\mathcal{R}(M)$), randomly selects $x_{i,2}$ consistently, and updates the chain. Otherwise, it returns a random consistent output.
 - On a backward query $(x_{i,1}, x_{i,2})$: $\mathcal{S}_i$ returns a random consistent x_{i-1} .

Indifferentiability Advantage Analysis. The indifferentiability advantage is defined as:

$$\mathbf{Adv}_{\text{Decohash}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) = |\Pr[\mathcal{D}^{\text{real}} \rightarrow 1] - \Pr[\mathcal{D}^{\text{ideal}} \rightarrow 1]|.$$

Assume $\mathcal{D}$ makes at most $q_i = q_i^+ + q_i^-$, $i \in [0, z]$ queries to the permutation oracles P_i (q_i^+ forward queries and q_i^- backward queries) and $q_{\mathcal{R}}$ queries to the random oracle. We define critical events:

- coll : Collision in H . Probability $\leq \mathbf{Adv}_H^{\text{coll}}(q_0)$.
- m-pre : H outputs a value in BadX (size p). Probability $\leq \mathbf{Adv}_H^{\text{m-pre}}(p, q_0)$.
- mitm : P_1^{-1} outputs a value that is a previous output of H . Probability $\leq Q_0 q_1 / 2^n$, where Q_0 stands for the number of H outputs after q_0 queries to P_0 and P_0^- .
- InvF : $\mathcal{D}$ correctly guesses a truncated part of r_i for some i , breaking the protection of F . Probability $\leq \sum_{i=1}^z q_i t / 2^{n-h}$.

Here, $t = \max\{t_1, t_2, \dots, t_z\}$ and t_i denotes the number of collisions on r_i among the $q_{\mathcal{R}}$ responses from $\mathcal{R}$. By this definition, no $(t_i + 1)$ collisions on r_i . Treating each r_i as an h -bit random value, t_i satisfies

$$\binom{q_{\mathcal{R}}}{t_i + 1} \left(\frac{1}{2^h}\right)^{t_i} < 1. \quad (1)$$

Using the inequality $\binom{q_{\mathcal{R}}}{t_i + 1} \leq \frac{q_{\mathcal{R}}^{t_i + 1}}{(t_i + 1)!}$, we solve for $q_{\mathcal{R}}$:

$$q_{\mathcal{R}} < (2^h)^{\frac{t_i}{t_i + 1}} \cdot ((t_i + 1)!)^{\frac{1}{t_i + 1}}.$$

Applying Stirling's approximation $((t_i + 1)!)^{\frac{1}{t_i + 1}} \leq \frac{t_i + 1}{e}$, we obtain $q_{\mathcal{R}} < \frac{t_i + 1}{e} \cdot (2^h)^{\frac{t_i}{t_i + 1}}$. This bound ensures that t_i -collisions on r_i may occur, while $(t_i + 1)$ -collisions are negligible. For such an r_i , there are t_i valid truncated parts that could lead to a correct preimage x_0 . With at most q_i backward queries to P_i^{-1} (where $q_i = q_i^+ + q_i^-$), the probability of success is at most $t_i q_i / 2^{n-h}$. Considering all z segments, the probability of event InvF is bounded by

$$\Pr[\text{InvF}] \leq \sum_{i=1}^z \frac{t_i q_i}{2^{n-h}} \leq \frac{z t q}{2^{n-h}}.$$

Let E be the event that no critical event occurs. Under E , the simulator perfectly mimics the real world, so the distinguisher's advantage is zero. Using the law of total probability and the triangle inequality, we bound the advantage:

$$\begin{aligned} \mathbf{Adv}_{\text{Decohash}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) &\leq \Pr[\neg E] \\ &\leq \Pr[\text{coll}] + \Pr[\text{m-pre}] + \Pr[\text{mitm}] + \Pr[\text{InvF}] \\ &\leq \mathbf{Adv}_H^{\text{coll}}(q_0) + \mathbf{Adv}_H^{\text{m-pre}}(p, q_0) + \frac{Q_0 q_1}{2^n} + \frac{z t q}{2^{n-h}}. \end{aligned}$$

Theorem 1. Let $H[P_0]$ be a collision-resistant and multiple-preimage-resistant hash function, and $F[P_1, \dots, P_z]$ be the truncating finalization defined above. Then the Decohash construction $F \circ H : \{0, 1\}^* \rightarrow \{0, 1\}^{zh}$ is indifferentiable from a random oracle. The indifferentiability advantage is bounded by:

$$\mathbf{Adv}_{\text{Decohash}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \mathbf{Adv}_H^{\text{coll}}(q_0) + \mathbf{Adv}_H^{\text{m-pre}}(p, q_0) + \frac{Q_0 q_1}{2^n} + \frac{z t q}{2^{n-h}},$$

where t is the maximum number of collision on r_i , and $q_{\mathcal{R}} < \frac{t+1}{e} \cdot (2^h)^{\frac{t}{t+1}}$ and $q_{\mathcal{S}_i} \leq q$.

The Decohash framework can be viewed as a *domain extender* for a fixed-input-length random oracle (FIL-RO), analogous to the FIL-RO domain extension paradigm established by Dodis, Ristenpart, and Shrimpton [19]. In their seminal work, they introduced the notion of *preimage awareness* (PrA)—a security property strictly between collision resistance and indistinguishability—and proved that the strengthened Merkle-Damgård (MD) transform preserves PrA. This preservation enables a modular construction of a variable-input-length (VIL) random oracle via the composition $F(x) = h(H(x))$, where h is a FIL-RO and H is a VIL-PrA hash function, achieving indistinguishability from a monolithic VIL-RO even when the underlying compression function is not itself indistinguishable from a FIL-RO.

Similarly, the Decohash framework realizes this domain extension through a structurally decoupled two-stage architecture: a variable-input-length component H ensuring collision and multiple-preimage resistance (functionally analogous to the PrA property required in the MD construction), followed by an independent fixed-input-length finalization phase F instantiated via independent random permutations. This separation ensures that the FIL component processes only pairwise-distinct outputs from the VIL compression phase, thereby eliminating the “randomness chain breakage” issues that plague homogeneous iterative designs. Consequently, analogous to the PrA-based MD construction, the Decohash framework effectively extends the domain of the underlying FIL primitive while providing rigorous security guarantees against length-extension attacks and achieving full indistinguishability from a random oracle.

The statement of Theorem 1 exhibits significant generality. Although it explicitly defines H as a hash function constructed from a public permutation P_0 , the core argument holds under a broader condition: H can be instantiated from *any* public primitive that is accessible to both the simulator and the distinguisher via queries. This primitive could be, for instance, a random function, a trapdoor function, or another permutation-based construction, provided the underlying query interface remains public.

However, the finalization step F plays a critical role in the security guarantee. The theorem specifically analyzes the case where F is implemented via output truncation, which is the most straightforward operation. While alternative constructions of F —such as those using a feedforward operation or other randomization techniques—may also achieve indistinguishability, each variant requires a dedicated and careful analysis. The security bounds and simulator strategies must be re-examined for any modification to F , as the finalization function directly controls the resistance against inversion attacks and mitm attacks.

Theorem 1 reveals a theoretical trade-off: by relaxing the security requirements of the iterative component to collision and multiple preimage resistance, and leveraging an independent randomized finalization, efficient hash constructions can achieve indistinguishability. This opens the possibility of optimizing H for speed while maintaining security through the randomized finalization step.

4 The Rocket Hash Constructions

4.1 Basic Idea

Building upon the Decohash framework, we apply its principles to design fast and secure hash constructions. The core design strategy mandates a strict separation between the VIL iterative compression phase and the FIL hash block generation phase, each employing independent permutations. The compression phase is optimized for high processing rate (speed) and needs only to provide collision resistance and multiple-preimage resistance, while the subsequent hash block generation phase, as defined in the Decohash framework, is fortified for strong security guarantees.

This section introduces two specific constructions named Rockets, reflecting their high speed and strong security. The first is based on the JH hash function, which was one of the five finalists in the NIST SHA-3 competition [42]. The second is a double-block wide construction derived from the standard Cipher-Block-Chaining (CBC) mode. Both constructions share a common three-phase structure:

1. **Initialization Phase:** This phase sets the initial value (IV) to the binary representation of the hash value length, and processes the IV through dedicated permutation rounds that establish the initial state for the subsequent iterative phase.
2. **Message Block Injection Phase:** This phase processes the input message blocks. To handle messages of arbitrary length, the input message is first subjected to a padding procedure. The design imposes no special requirements on the padding method; any standard scheme that ensures the original message can be uniquely recovered from the padded message is acceptable. A common approach is to append a single bit '1' followed by the minimum number of '0' bits (possibly none) required to make the total length of the padded message a multiple of $n/2$ for Rocket-JH, and a multiple of n for Rocket-DoubleCBC.
3. **Hash Block Generation Phase:** This final FIL component follows the DecoHash framework to generate hash blocks (with $h = n/2$).

Pairwise Distinct Round Permutations. A distinctive feature of our proposed Rocket hash functions is the *requirement for pairwise distinct permutations across all rounds*. Unlike conventional iterative designs that employ a single round function (or a small set of keyed permutations with varying constants), our construction mandates that each processing round utilizes an *independent* random permutation.

This architectural choice reflects a deliberate design philosophy emphasizing *modular security decomposition* and serves multiple critical objectives. First, it eliminates problematic query cycles between the VIL and FIL components that could otherwise arise from adversarial exploitation of permutation structural similarities. By ensuring that the permutation interface exposed to the adversary remains independent across rounds, the simulator can maintain precise query state isolation, thereby facilitating tight indistinguishability proofs with manageable complexity. Second, it ensures statistical independence across the iterative compression chain, preventing adversaries from leveraging algebraic relationships between consecutive permutations to mount meet-in-the-middle or preimage amplification attacks. Third, the independence requirement enforces strict *knowledge compartmentalization*: information about internal states or permutation queries in one round provides no advantage for predicting or influencing subsequent rounds. This compartmentalization not only simplifies the security analysis but yields a crucial quantitative benefit—the security bounds depend linearly on the message length l rather than quadratically as l^2 , which is typical in homogeneous designs where the reuse of a single primitive creates cross-round dependencies.

Finally, this paradigm of pairwise distinct round functions establishes a versatile template that transcends the specific Rocket instantiations. By demonstrating that cryptographic primitives need not be monolithic but can be composed from independent, functionally specialized modules, we inspire further constructions within the Decohash framework and beyond, encouraging future designs that trade structural redundancy for enhanced security assurance and analytical clarity.

Practical Instantiation. While generating a sequence of truly independent permutations appears challenging for practical implementation, this requirement is efficiently realized through the CTR-Perm domain-separation technique (detailed in Section 5), which derives effectively independent function instances from a single underlying large permutation via counter-based input diversification.

4.2 Rocket-JH

Illustration and Specification Rocket-JH is constructed upon the JH iteration structure [42], utilizing pairwise distinct round permutations $P_{i,1}$ ($i \in [0, l]$) and $P_{j,3}$ ($j \in [1, z]$) on $\{0, 1\}^n$. Each state variable has a size of $n/2$ bits. The detail procedures are illustrated in Fig. 2 and specified in Alg. 1.

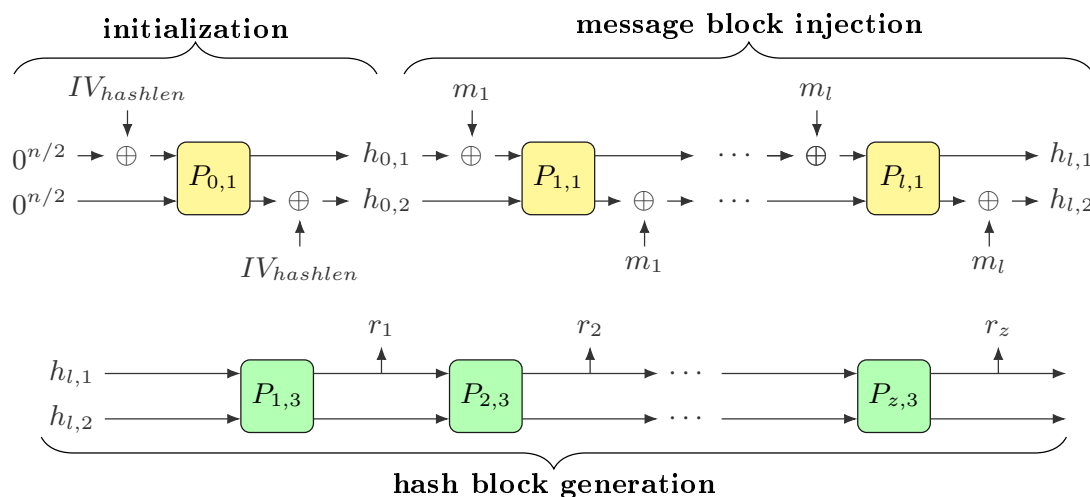


Fig. 2. Rocket-JH: Three-Phase Construction Overview

Algorithm 1 Rocket-JH specification

- 1: on input a message $M \in \{0, 1\}^*$ and a hash value length $hashlen$
 - 2: *Initialization Phase*
 - 3: $h_{0,1} || h_{0,2} = P_{0,1}(IV_{hashlen} || 0^{n/2}) \oplus (0^{n/2} || IV_{hashlen})$, with $|IV_{hashlen}| = n/2$
 - 4: *Message Block Injection Phase*
 - 5: $m_1 m_2 \dots m_l \leftarrow \mathbf{Pad}(M)$, with $|m_i| = n/2$ for $i \in [1, l]$
 - 6: **for each** $i \in [1, l]$ **do**
 - 7: $x_{i,1} = h_{i-1,1} \oplus m_i$, $x_{i,2} = h_{i-1,2}$
 - 8: $y_{i,1} || y_{i,2} = P_{i,1}(x_{i,1} || x_{i,2})$
 - 9: $h_{i,1} = y_{i,1}$, $h_{i,2} = y_{i,2} \oplus m_i$
 - 10: **end for**
 - 11: *Hash Block Generation Phase*
 - 12: $z = \lceil 2(hashlen)/n \rceil$
 - 13: **for each** $i \in [1, z]$ **do**
 - 14: $h_{l+i,1} || h_{l+i,2} = P_{i,3}(h_{l+i-1,1} || h_{l+i-1,2})$
 - 15: $r_i = h_{l+i,1}$
 - 16: **end for**
 - 17: $hash = \mathbf{LMB}(hashlen, r_1 || r_2 || \dots || r_z)$
 - 18: **return** $hash$
-

Internal State Propagation Let R1 be the Rocket-JH construction excluding the hash block generation phase:

$$R1[P_{i,1}](M) = h_{l,1} || h_{l,2}, \text{ with } i \in [0, l].$$

Since the permutations $P_{i,1}$ are public, adversaries can make arbitrary queries to $P_{i,1}$ and $P_{i,1}^{-1}$. Let q_i^+ , q_i^- , and $q_i = q_i^+ + q_i^-$ denote the number of forward, backward, and total queries in round i , respectively, and let $q = \max_i q_i$.

In round $i \in [1, l]$, assume an adversary $\mathcal{A}$ has obtained q internal states $h_{i-1,1} \| h_{i-1,2}$. Let t be the maximum multiplicity of any colliding half-state value $h_{i-1,2}$ among these q states. Let $h_{i-1,2}^*$ denote such a value that occurs t times, and let the corresponding distinct half-states be $h_{i-1,1}^{(1)}, \dots, h_{i-1,1}^{(t)}$.

If the adversary queries $P_{i,1}$ at a point $x_{i,1}^* \| h_{i-1,2}^*$ for some chosen $x_{i,1}^*$, they receive the output $y_{i,1}^* \| y_{i,2}^*$. This single query allows the construction of up to t distinct input-output tuples:

$$(h_{i-1,1}^{(j)}, h_{i-1,2}^*, m_i^{(j)}, h_{i,1}^*, h_{i,2}^{(j)})_{j \in [1, t]},$$

where

$$m_i^{(j)} = h_{i-1,1}^{(j)} \oplus x_{i,1}^*, \quad h_{i,1}^* = y_{i,1}^*, \quad h_{i,2}^{(j)} = y_{i,2}^* \oplus m_i^{(j)}.$$

The resulting states $(h_{i,1}^{(j)}, h_{i,2}^{(j)})_{j \in [1, t]}$ are pairwise distinct. Hence, via a single forward query to $P_{i,1}$, the adversary can obtain up to t new internal states at round i . Over the course of $q_i^+ \leq q$ forward queries, $\mathcal{A}$ can therefore acquire at most tq distinct states of the form $h_{i,1} \| h_{i,2}$.

Analysis of the parameter t Notice that t is the maximum collision multiplicity for half-state values (h_1 or h_2) among q internal states, excluding trivial collisions arising from identical message prefixes or suffixes. This parameter directly influences the efficiency of internal states propagation, as each collision on a half-state provides an additional candidate path for the adversary to exploit. Assuming states are uniformly random, we require that the expected number of $(t+1)$ -collisions be less than one to ensure that such events remain statistically negligible:

$$\binom{q}{t+1} \left(\frac{1}{2^{n/2}} \right)^t < 1.$$

In subsequent collision analysis, we require $q \leq 2^{3n/8}$. For this range of q , the condition above yields $t \leq 3$ as the maximum plausible collision multiplicity.

Preimage Resistance of R1 Given a challenge (h_1, h_2) , an adversary $\mathcal{A}$ attempts to find a message M such that $R1(M) = h_1 \| h_2$. For practical hash functions, the number of valid output lengths, and thus the number of possible initial values IV_{hashlen} , is limited. Let k denote this number of valid initial states $h_{0,1} \| h_{0,2} = P_{0,1}(IV_{\text{hashlen}} \| 0^{n/2}) \oplus (0^{n/2} \| IV_{\text{hashlen}})$. Both the initial state and the target state (h_1, h_2) are fixed, so $\mathcal{A}$ must link them through R1 using one of three strategies:

1. **Forward-only strategy:** $\mathcal{A}$ propagates states from one of the k valid initial states toward the target. For each round i , with at most t collisions on $h_{i-1,2}$, a single query to $P_{i,1}$ yields up to t distinct states at round i . With $q_i^+ \leq q$ queries in each round and over all l rounds, the probability of reaching (h_1, h_2) is upper bounded by:

$$\Pr[\text{forward reach}] \leq \sum_{i=1}^l \frac{tq}{2^n} = \frac{tql}{2^n}.$$

2. **Backward-only strategy:** $\mathcal{A}$ works backward from the target (h_1, h_2) toward one of the k valid initial states. Due to symmetry in R1, the analysis mirrors the forward strategy. The success probability is dominated by matching one of the k possible initial states, which gives:

$$\Pr[\text{backward reach}] \leq \frac{ktq}{2^n}.$$

3. **Meet-in-the-middle strategy:** $\mathcal{A}$ combines forward queries from an initial state to an intermediate round i with backward queries from the target to the same intermediate round i . With at most tq_i^+ states from the forward direction and tq_{i+1}^- states from the backward direction, the probability of a match at round i is:

$$\Pr[\text{mitm at round } i] \leq \binom{tq_i^+}{1} \binom{tq_{i+1}^-}{1} \frac{1}{2^n} \leq \frac{t^2 q^2}{2^n}.$$

Summing over all possible intermediate rounds gives the overall bound:

$$\Pr[\text{mitm}] \leq \sum_{i=1}^l \frac{t^2 q^2}{2^n} = \frac{t^2 q^2 l}{2^n}.$$

The meet-in-the-middle strategy dominates, so the preimage advantage is:

$$\mathbf{Adv}_{\mathbf{R1}}^{\text{pre}}(q) \leq \frac{t^2 q^2 l}{2^n}.$$

Multiple-Preimage Resistance of R1 Given p challenges $(h_1^{(j)}, h_2^{(j)})$ for $j \in [1, p]$, an adversary $\mathcal{A}$ attempts to find a message M such that $\exists j \in [1, p]$ with $R1(M) = h_1^{(j)} \| h_2^{(j)}$. The k valid initial states are fixed, and $\mathcal{A}$ must link one of them to one of the p targets through R1.

A naive approach is to repeat the preimage attack p times, which gives the bound:

$$\mathbf{Adv}_{\mathbf{R1}}^{\text{m-pre}}(p, q) \leq p \times \mathbf{Adv}_{\mathbf{R1}}^{\text{pre}}(q).$$

For the forward-only strategy, the adversary can succeed by matching any one of the p targets. This yields:

$$\Pr[\text{forward reach}] \leq \sum_{i=1}^l \sum_{j=1}^p \frac{tq}{2^n} = \frac{tpql}{2^n} \leq \frac{tq^2 l}{2^n} \quad (\text{when } p \leq q).$$

For the backward-only and meet-in-the-middle strategies, the total query budget remains q . Since these strategies rely on generating a set of candidate states to match a specific target, the adversary's optimal approach is to focus all queries on the most promising single target. Consequently, the advantage is not significantly increased by having multiple targets and is bounded as in the single-target case.

Thus, the multiple-preimage advantage is:

$$\mathbf{Adv}_{\mathbf{R1}}^{\text{m-pre}}(q) = \mathbf{Adv}_{\mathbf{R1}}^{\text{pre}}(q) \leq \frac{9q^2 l}{2^n}, \quad (\text{for } p \leq q, \ q \leq 2^{3n/8}, \text{ and } t = 3).$$

Collision Resistance of R1 An adversary $\mathcal{A}$ aims to find two distinct messages $M \neq M'$ such that $R1(M) = R1(M')$. Let $m_1 \cdots m_i = \mathbf{Pad}(M)$ and $m'_1 \cdots m'_{i'} = \mathbf{Pad}(M')$ be the padded message blocks. We consider two cases based on the message lengths.

Case 1: Different Message Lengths ($i \neq i'$). Without loss of generality, assume $i' < i$. After processing i' blocks, M' reaches its final state $h'_{i',1} \| h'_{i',2}$ deterministically. To achieve a collision, $\mathcal{A}$ must connect $h_{i',1} \| h_{i',2}$ (the state of M after i' rounds) to this final state via the remaining $i - i'$ rounds of R1. This reduces to a preimage attack on a shortened version of R1 with $i - i'$ rounds. Applying the meet-in-the-middle bound from preimage analysis:

$$\Pr[\text{collision} \mid i \neq i'] \leq \frac{t^2 q^2 (i - i')}{2^n} < \frac{t^2 q^2 l}{2^n}.$$

Case 2: Equal Message Lengths ($i = i'$). Let i denote the common length. We analyze the probability of collision at each round, starting from $i = 0$ up to $i = l$.

For round $i = 0$, there are k possible initial states $(h_{0,1}, h_{0,2})$ determined by the IV_{hashlen} . The probability that two distinct initial states collide is:

$$\Pr[\text{collision at round } 0] = \Pr[\exists h_{0,1} \| h_{0,2} = h'_{0,1} \| h'_{0,2}] \leq \binom{k}{2} \frac{1}{2^n},$$

which is negligible.

For each subsequent round $i \in [1, l]$, we condition on the event that no earlier collision has occurred. A collision at round i occurs when $h_{i,1} \| h_{i,2} = h'_{i,1} \| h'_{i,2}$ while $h_{i-1,1} \| h_{i-1,2} \neq h'_{i-1,1} \| h'_{i-1,2}$. From the specification of Rocket-JH, this equality can be expressed as:

$$P_{i,1}(x_{i,1} \| x_{i,2}) \oplus P_{i,1}(x'_{i,1} \| x'_{i,2}) = 0^{n/2} \| (m_i \oplus m'_i)$$

where $x_{i,1} = h_{i-1,1} \oplus m_i$, $x_{i,2} = h_{i-1,2}$, and similarly for the primed variables.

If $m_i = m'_i$, then the equation simplifies to $P_{i,1}(x_{i,1} \| x_{i,2}) = P_{i,1}(x'_{i,1} \| x'_{i,2})$, which implies $x_{i,1} \| x_{i,2} = x'_{i,1} \| x'_{i,2}$ since $P_{i,1}$ is a permutation. This further yields $h_{i-1,1} \| h_{i-1,2} = h'_{i-1,1} \| h'_{i-1,2}$, contradicting our assumption of no earlier collision. Thus, any non-trivial collision requires $m_i \neq m'_i$.

To achieve such a collision, the adversary must find two distinct inputs to $P_{i,1}$ whose outputs differ by $0^{n/2} \| \Delta$ for some non-zero $\Delta = m_i \oplus m'_i$. The adversary can make at most q queries to $P_{i,1}$ and its inverse in each round. We bound the probability for round i based on the direction of the two queries:

- Both queries forward: The adversary makes q_i^+ forward queries to $P_{i,1}$. By the birthday bound, the probability of finding two distinct inputs that produce outputs satisfying the required difference is:

$$\Pr[\text{collision at round } i \mid \text{FF}] \leq \binom{tq_i^+}{2} \cdot \frac{1}{2^n} \leq \frac{t^2 q^2}{2^{n+1}}.$$

- Both queries backward: The adversary proceeds as follows. First, $\mathcal{A}$ selects and fixes any value $h_{i,1}^*$. For this fixed value, $\mathcal{A}$ makes q_i^- inverse queries to $P_{i,1}^{-1}$, each of the form $h_{i,1}^* \| y_{i,2}$ (with $y_{i,2}$ varying), and receives the corresponding outputs $x_{i,1} \| x_{i,2}$. Meanwhile, the adversary has at most tq_{i-1}^+ forward-propagated states from round $i-1$. To construct a collision, $\mathcal{A}$ selects any two distinct states from these forward states and any two distinct outputs from the backward queries. The probability that the first forward-backward pair satisfies $h_{i-1,2} = x_{i,2}$ and the second pair satisfies $h'_{i-1,2} = x'_{i,2}$ is bounded by $\binom{tq_{i-1}^+}{2} \binom{q_i^-}{2} \left(\frac{1}{2^{n/2}}\right)^2$, as each equality condition on the $n/2$ -bit values holds with probability $1/2^{n/2}$.

For such matching pairs, $\mathcal{A}$ computes the corresponding message blocks as $m_i = h_{i-1,1} \oplus x_{i,1}$ and $m'_i = h'_{i-1,1} \oplus x'_{i,1}$. The probability that these two message blocks yield a collision at round i (i.e., $y_{i,2} \oplus m_i = y'_{i,2} \oplus m'_i$) is $\frac{1}{2^{n/2}}$, since the $h_{i,1}$ values are fixed to $h_{i,1}^*$ and the collision depends solely on the $n/2$ -bit $h_{i,2}$ components.

Thus, the overall probability of a collision at round i using this meet-in-the-middle strategy is:

$$\Pr[\text{collision at round } i \mid \text{BB}] \leq \binom{tq_{i-1}^+}{2} \binom{q_i^-}{2} \left(\frac{1}{2^{n/2}}\right)^3 \leq \frac{t^2 q^4}{2^{3n/2+2}}.$$

- Forward-then-backward: The backward query must link the final state at round i produced by the forward query back to one of the internal states at round $i - 1$ (of which there are at most tq_{i-1}^+). This gives:

$$\Pr[\text{collision at round } i \mid \text{FtB}] \leq q_i^- \times \frac{tq_{i-1}^+}{2^n} \leq \frac{tq^2}{2^n},$$

which is always smaller than $\frac{t^2q^2}{2^{n+1}}$ from the FF case.

- Backward-then-forward: The adversary first makes q_i^- backward queries, trying to invert any final state at round i to one of the forward-propagated states (at most tq_{i-1}^+), and then it makes q_i^+ forward queries, ensure the forward computation can match the inverted final state. The probability of this two-step match is:

$$\Pr[\text{collision at round } i \mid \text{BtF}] \leq \binom{q_i^-}{1} \binom{tq_{i-1}^+}{1} \frac{1}{2^n} \times \binom{q_i^+}{1} \frac{1}{2^n} \leq \frac{tq^3}{2^{2n}},$$

which is always smaller than $\frac{t^2q^4}{2^{3n/2+2}}$ from the BB case.

Summing this probability over all l rounds $i \in [1, l]$, the total probability of collision for equal-length messages (ignoring the negligible round-0 term) is:

$$\sum_{i=1}^l \left(\frac{t^2q^2}{2^{n+1}} + \frac{t^2q^4}{2^{3n/2+2}} \right) = \frac{t^2q^2l}{2^{n+1}} + \frac{t^2q^4l}{2^{3n/2+2}}.$$

Overall Collision Bound. Combining both cases (different lengths and equal lengths), the overall collision probability is bounded by the sum of the dominant terms. Therefore, the total advantage is:

$$\mathbf{Adv}_{\text{R1}}^{\text{coll}}(q) \leq \frac{t^2q^2l}{2^n} + \frac{t^2q^4l}{2^{3n/2+2}}.$$

Substituting the maximum collision multiplicity $t = 3$ (valid for $q \leq 2^{3n/8}$) yields the final collision resistance bound:

$$\mathbf{Adv}_{\text{R1}}^{\text{coll}}(q) \leq \frac{9q^2l}{2^n} + \frac{9q^4l}{2^{3n/2+2}}, \quad \text{for } q \leq 2^{3n/8}.$$

Indifferentiability of Rocket-JH Applying Theorem 1 (with $Q_0 \leq ql$ and $q_i \leq q$ for $i \in [1, z]$), we conclude that Rocket-JH is indifferentiable from a random function. Specifically, we have

$$\mathbf{Adv}_{\text{Rocket-JH}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{19q^2l}{2^n} + \frac{9q^4l}{2^{3n/2+2}} + \frac{3zq}{2^{n/2}},$$

with $q \leq 2^{3n/8}$, $t \leq 3$ and $h = n/2$.

4.3 Rocket-DoubleCBC

Illustration and Specification To further enhance the efficiency of the hash iteration process, we propose Rocket-DoubleCBC, a construction that employs two independent cipher-block-chaining (CBC) lanes operating in parallel. The design remains based on pairwise distinct random permutations. This structure is particularly suited for multi-core platforms, where it can achieve significantly high throughput. As depicted in Fig. 3 and detailed in Algorithm 2, before the hash block generation phase, all variables are n bits in length.

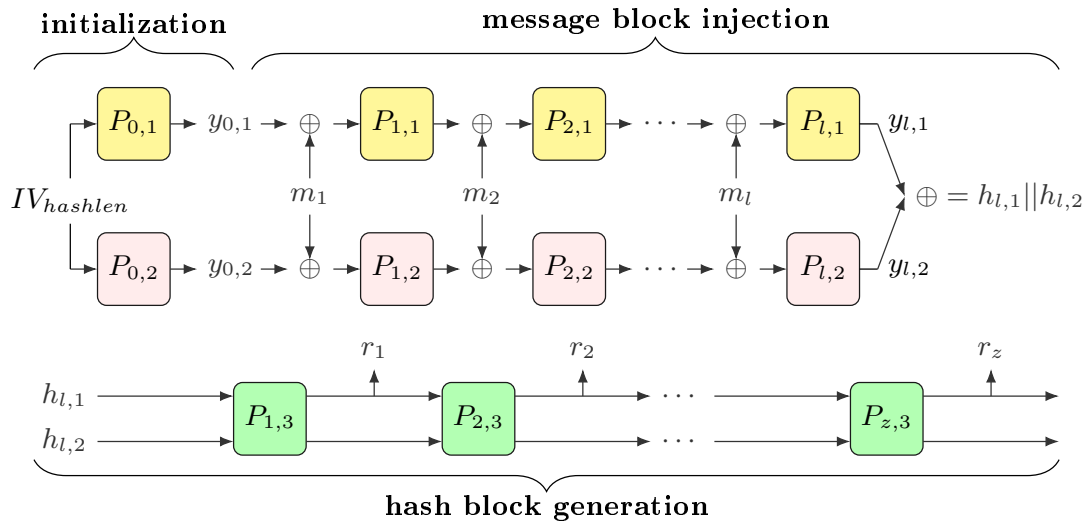


Fig. 3. Rocket-DoubleCBC: Three-Phase Construction Overview

Algorithm 2 Rocket-DoubleCBC specification

- 1: on input a message $M \in \{0,1\}^*$ and a hash value length $hashlen$
 - 2:
 - 3: *Initialization Phase*
 - 4: $y_{0,1} = P_{0,1}(IV_{hashlen})$, $y_{0,2} = P_{0,2}(IV_{hashlen})$ with $|IV_{hashlen}| = n$
 - 5:
 - 6: *Message Block Injection Phase*
 - 7: $m_1 m_2 \dots m_l \leftarrow \mathbf{Pad}(M)$, with $|m_i| = n$ for $i \in [1, l]$
 - 8: **for each** $i \in [1, l]$ **do**
 - 9: $x_{i,1} = y_{i-1,1} \oplus m_i$, $x_{i,2} = y_{i-1,2} \oplus m_i$
 - 10: $y_{i,1} = P_{i,1}(x_{i,1})$, $y_{i,2} = P_{i,2}(x_{i,2})$
 - 11: **end for**
 - 12: $h_{l,1} || h_{l,2} = y_{l,1} \oplus y_{l,2}$
 - 13:
 - 14: do the same *Hash Block Generation Phase* as in Alg. 1
-

Internal State Propagation Let R2 denote the Rocket-DoubleCBC construction without the hash block generation phase, defined as:

$$R2(M) = y_{l,1} \oplus y_{l,2}.$$

To analyze the hash security properties (collision, preimage, and multiple-preimage resistance) of R2, we consider an adversary $\mathcal{A}$ that can make forward and backward queries to the underlying permutations $P_{i,1}$ and $P_{i,2}$ for $i \in [0, l]$, and use the responses to construct internal states of R2.

Validity of Internal States. According to the specification of R2, a valid internal state must be computed sequentially: given a previous valid state $(y_{i-1,1}, y_{i-1,2})$, one selects a message block m_i and computes forward queries $x_{i,1} = y_{i-1,1} \oplus m_i$ to $P_{i,1}$ and $x_{i,2} = y_{i-1,2} \oplus m_i$ to $P_{i,2}$ to obtain the next state $(y_{i,1}, y_{i,2})$. An essential invariant of this construction is that the XOR of the two round permutation inputs remains equal to the XOR of the two state components:

$$x_{i,1} \oplus x_{i,2} = y_{i-1,1} \oplus y_{i-1,2}.$$

Consequently, if a pair $(y_{i-1,1}, y_{i-1,2})$ is constructed arbitrarily (i.e., not as the output of a legitimate computation), the resulting state $(y_{i,1}, y_{i,2})$ will almost certainly be *invalid* for R2. This validity condition significantly restricts the adversary's ability to generate new states through arbitrary queries.

State Propagation Efficiency. Given q valid internal states at round $i - 1$, and restricting the adversary to at most q (forward and backward) queries each to $P_{i,1}$ and $P_{i,2}$, how many valid states at round i can be obtained? Let t denote this number.

A trivial upper bound is $t \leq q^2$, as q forward queries to each permutation can produce at most q^2 pairs $(x_{i,1}, x_{i,2})$. However, the validity condition $x_{i,1} \oplus x_{i,2} = y_{i-1,1} \oplus y_{i-1,2}$ substantially reduces this number. Conversely, a trivial lower bound is $t \geq q$, achieved by fixing any valid state $(y_{i-1,1}, y_{i-1,2})$ and generating q pairs $(x_{i,1}^{(j)}, x_{i,2}^{(j)})$ where $x_{i,1}^{(j)}$ are distinct and $x_{i,2}^{(j)} = x_{i,1}^{(j)} \oplus y_{i-1,1} \oplus y_{i-1,2}$.

More sophisticated strategies can improve this lower bound. Consider the following approach: Given q valid states $(y_{i-1,1}^{(j)}, y_{i-1,2}^{(j)})$ for $j \in [1, q]$, let $\Delta_{i-1}^{(j)} = y_{i-1,1}^{(j)} \oplus y_{i-1,2}^{(j)}$. The adversary proceeds as follows:

1. *Query $P_{i,1}$:* Select a constant $\text{Cst} \in \{0, 1\}^n$ and query $x_{i,1}^{(j)} = \Delta_{i-1}^{(j)} \oplus \text{Cst}$ for all $j \in [1, q]$.
2. *First query to $P_{i,2}$:* Query $x_{i,2}^{(1)} = \text{Cst}$. Since $x_{i,2}^{(1)} = x_{i,1}^{(1)} \oplus \Delta_{i-1}^{(1)}$ for all j , this creates q valid pairs of the form $(x_{i,1}^{(j)}, \text{Cst})$.
3. *Subsequent queries to $P_{i,2}$:* For $j \in [2, q]$, query $x_{i,2}^{(j)} = x_{i,1}^{(1)} \oplus \Delta_{i-1}^{(j)}$. This immediately creates $q - 1$ valid pairs $(x_{i,1}^{(1)}, x_{i,2}^{(j)})$. Moreover, for each $j \geq 2$, we have another representation $x_{i,2}^{(j)} = x_{i,1}^{(j)} \oplus \Delta_{i-1}^{(1)}$, yielding an additional $q - 1$ valid pairs $(x_{i,1}^{(j)}, x_{i,2}^{(j)})$.

In total, this method produces $q + 2(q - 1) < 3q$ valid pairs for $P_{i,1}$ and $P_{i,2}$. More intensive analysis shows that, given $O(q)$ valid internal states at round $i - 1$ and restricting the adversary to at most q queries each to $P_{i,1}$ and $P_{i,2}$, the R2 iteration structure limits the number of valid states at round i to

$$\leq (3/(2\epsilon) + 1)q = \Theta(q), \quad \text{for } q < 2^{n/2 - \epsilon n} \text{ and } \epsilon = \Theta(1).$$

A rigorous proof of this bound is presented in Appendix A. For our subsequent security analysis, we will choose $\epsilon = 1/20$, and use the concrete bound $t < 32 = 2^5$ for all $q < 2^{0.45n}$.

Preimage and Multiple-Preimage Resistance of R2 The preimage and multiple-preimage resistance analysis for R2 follows the same framework as established for R1 in Section 4.2. Due to the similar structure of the state propagation and the same bounds on valid internal states (with parameter t), we obtain identical asymptotic security bounds.

Specifically, for any adversary $\mathcal{A}$ making at most q queries to each round permutation, the advantage in finding a preimage for a target value Δ_y is bounded by:

$$\text{Adv}_{\text{R2}}^{\text{pre}}(q) \leq \frac{t^2 q^2 l}{2^n}.$$

Similarly, for multiple-preimage resistance with $p \leq q$ challenge targets, the advantage satisfies:

$$\text{Adv}_{\text{R2}}^{\text{m-pre}}(p, q) = \text{Adv}_{\text{R2}}^{\text{pre}}(q) \leq \frac{t^2 q^2 l}{2^n}.$$

The detailed derivation of these bounds, including the forward-only, backward-only, and meet-in-the-middle strategies, is provided in Appendix B.

Collision Resistance of R2 For the R2 construction with at most l rounds, an adversary $\mathcal{A}$ aims to find two distinct messages $M \neq M'$ such that $\text{R2}(M) = \text{R2}(M')$. Let $m_1 \cdots m_i = \text{Pad}(M)$ and $m'_1 \cdots m'_{i'} = \text{Pad}(M')$. The collision analysis considers two cases based on message lengths.

Case 1: Different Message Lengths ($i' < i$). Since the round permutations are pairwise distinct, $\text{R2}(M')$ is determined after i' rounds. The internal state $(y_{i',1}, y_{i',2})$ is fixed. To achieve a collision, $\mathcal{A}$ must link this state to $\text{R2}(M) = \text{R2}(M')$ over the remaining $i - i'$ rounds. This reduces to finding a preimage for a fixed target on a shortened $(i - i')$ -round instance of R2. Applying the meet-in-the-middle bound from preimage analysis:

$$\Pr[\text{collision} \mid i' < i] \leq \frac{t^2 q^2 (i - i')}{2^n} < \frac{t^2 q^2 l}{2^n}.$$

Case 2: Equal Message Lengths ($i' = i$). Let i denote the common length. We analyze the probability of collision at each round, starting from $i = 0$ up to $i = l$.

For round $i = 0$, there are k possible initial pairs $(y_{0,1}, y_{0,2})$ determined by IV_{hashlen} . The probability that two distinct initial state differences collide is:

$$\Pr[\text{collision at round } 0] = \Pr[\exists y_{0,1} \oplus y_{0,2} = y'_{0,1} \oplus y'_{0,2}] \leq \binom{k}{2} \frac{1}{2^n},$$

which is negligible for practical parameters.

For each subsequent round $i \in [1, l]$, we condition on the event that no earlier collision has occurred. A collision at round i occurs when $y_{i,1} \oplus y_{i,2} = y'_{i,1} \oplus y'_{i,2}$ while $y_{i-1,1} \oplus y_{i-1,2} \neq y'_{i-1,1} \oplus y'_{i-1,2}$. From the R2 specification:

$$y_{i,1} \oplus y_{i,2} = P_{i,1}(y_{i-1,1} \oplus m_i) \oplus P_{i,2}(y_{i-1,2} \oplus m_i).$$

The collision $y_{i,1} \oplus y_{i,2} = y'_{i,1} \oplus y'_{i,2}$ implies that the XOR of the two permutation outputs must be equal. That is:

$$P_{i,1}(y_{i-1,1} \oplus m_i) \oplus P_{i,2}(y_{i-1,2} \oplus m_i) = P_{i,1}(y'_{i-1,1} \oplus m'_i) \oplus P_{i,2}(y'_{i-1,2} \oplus m'_i).$$

This can be rewritten as:

$$P_{i,1}(y_{i-1,1} \oplus m_i) \oplus P_{i,1}(y'_{i-1,1} \oplus m'_i) = P_{i,2}(y_{i-1,2} \oplus m_i) \oplus P_{i,2}(y'_{i-1,2} \oplus m'_i).$$

If $y_{i-1,1} \oplus m_i = y'_{i-1,1} \oplus m'_i$ and $y_{i-1,2} \oplus m_i = y'_{i-1,2} \oplus m'_i$ both hold, then we would have $y_{i-1,1} \oplus y_{i-1,2} = y'_{i-1,1} \oplus y'_{i-1,2}$, contradicting our assumption. Therefore, the two input pairs to the permutations must be distinct for a non-trivial collision.

To achieve such a collision, the adversary must find two distinct input pairs whose outputs satisfy the above equality. The adversary can make at most q queries to each permutation and its inverse per round. We bound the probability for round i based on the direction of the queries:

- Both forward queries: The adversary makes $q_{i,1}^+$ and $q_{i,2}^+$ forward queries to $P_{i,1}$ and $P_{i,2}$ respectively. With at most tq valid previous states, the adversary can generate at most tq candidate pairs $(y_{i,1}, y_{i,2})$. By the birthday bound, the probability that any two such pairs produce a collision $y_{i,1} \oplus y_{i,2} = y'_{i,1} \oplus y'_{i,2}$ is:

$$\Pr[\text{collision at round } i \mid \text{FF}] \leq \binom{tq}{2} \cdot \frac{1}{2^n} \leq \frac{t^2 q^2}{2^{n+1}}.$$

- Both backward queries: The adversary fixes a target value Δ_i^* and makes at most $q_{i,1}^-$ inverse queries $y_{i,1}^{(j)}$ to $P_{i,1}^{-1}$ to obtain inputs $x_{i,1}^{(j)}$, and at most $q_{i,2}^-$ inverse queries to $P_{i,2}^{-1}$ on values $y_{i,2}^{(j)} = y_{i,1}^{(j)} \oplus \Delta_i^*$ to obtain inputs $x_{i,2}^{(j)}$. Using a meet-in-the-middle approach, the adversary must match two backward-query output pairs $(x_{i,1}, x_{i,2})$ and $(x'_{i,1}, x'_{i,2})$ with two forward-propagated states $(y_{i-1,1}, y_{i-1,2})$ and $(y'_{i-1,1}, y'_{i-1,2})$ from round $i-1$ (at most tq possibilities each). This requires two n -bit XOR equations $x_{i,1} \oplus x_{i,2} = y_{i-1,1} \oplus y_{i-1,2}$ and $x'_{i,1} \oplus x'_{i,2} = y'_{i-1,1} \oplus y'_{i-1,2}$, yielding:

$$\Pr[\text{collision at round } i \mid \text{BB}] \leq \binom{tq}{2} \binom{q_{i,1}^-}{2} \left(\frac{1}{2^n}\right)^2 \leq \frac{t^2 q^4}{2^{2n+2}}.$$

- Forward-then-backward: Given tq valid internal states from round $i-1$, the adversary makes forward queries at round i to obtain a valid internal state $(y_{i,1}^*, y_{i,2}^*)$. For this specific state, the adversary then makes at most $q_{i,1}^-$ backward queries $y_{i,1}^{(j)}$ to $P_{i,1}^{-1}$ and corresponding queries $y_{i,2}^{(j)} = y_{i,1}^{(j)} \oplus y_{i,1}^* \oplus y_{i,2}^*$ to $P_{i,2}^{-1}$. The adversary must match at least one resulting pair $(x_{i,1}^{(j)}, x_{i,2}^{(j)})$ against one of the tq forward state differences from round $i-1$. This gives:

$$\Pr[\text{collision at round } i \mid \text{FtB}] \leq q_{i,1}^- \cdot \frac{tq}{2^n} \leq \frac{tq^2}{2^n}.$$

- Backward-then-forward: The adversary first makes at most $q_{i,1}^-$ inverse queries to $P_{i,1}^{-1}$ and $P_{i,2}^{-1}$, attempting to match one of the resulting input pairs against the tq forward state differences from round $i-1$. Upon a successful match, the adversary then makes at most $q_{i,1}^+$ forward queries to $P_{i,1}$ and $P_{i,2}$ to complete the collision. The probability of this two-step match is:

$$\Pr[\text{collision at round } i \mid \text{BtF}] \leq \binom{q_{i,1}^-}{1} \frac{tq}{2^n} \cdot \binom{q_{i,1}^+}{1} \frac{1}{2^n} \leq \frac{tq^3}{2^{2n}}.$$

For $q \leq 2^{0.45n}$, the dominant term among these four strategies for $i \geq 1$ is the FF case, giving $\frac{t^2 q^2}{2^{n+1}}$. Summing the dominant term over all l rounds, the total probability for equal-length messages is:

$$\Pr[\text{collision} \mid i' = i] \leq \sum_{i=1}^l \frac{t^2 q^2}{2^{n+1}} = \frac{t^2 q^2 l}{2^{n+1}}.$$

Overall Collision Bound. Combining both cases, the overall collision probability is dominated by the unequal-length case. The negligible round-0 term is absorbed into the main bound. Therefore:

$$\mathbf{Adv}_{\text{R2}}^{\text{coll}}(q) \leq \frac{t^2 q^2 l}{2^n}.$$

Indifferentiability of Rocket-DoubleCBC Applying Theorem 1 and combining our multiple-preimage and collision resistance bounds, we conclude that Rocket-DoubleCBC is indifferentiable from a random oracle. The indifferentiability advantage of an adversary making at most q queries to each underlying permutation is bounded by:

$$\mathbf{Adv}_{\text{Rocket-DoubleCBC}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{t^2 q^2 l}{2^{n-1}} + \frac{ql \cdot q}{2^n} + \frac{zt'q}{2^{n/2}}.$$

Substituting the concrete bounds $t < 32$ (from state propagation analysis) and $t' \leq 9$ (derived from Eq. (1) with $q_{\mathcal{R}} < 2^{0.45n}$ and $h = n/2$ in Section 3.2) yields the simplified expression:

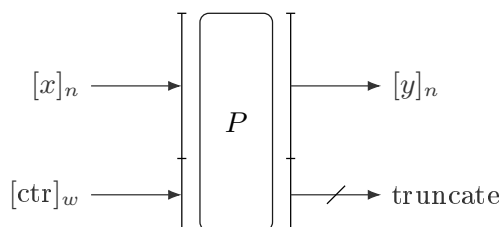
$$\mathbf{Adv}_{\text{Rocket-DoubleCBC}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{q^2 l}{2^{n-11}} + \frac{9zq}{2^{n/2}}, \quad \text{for } q < 2^{0.45n}.$$

5 CTR-Perm: Domain Separation via Counter Mode

5.1 Design Motivation: The Independence Problem

The Rocket hash constructions require 2^w *pairwise independent* round permutations to achieve provable indifferenciability. However, large public permutations (e.g., Keccak- p [1600]) lack a key schedule—each permutation provides only a single functional instance. A naive approach using round constants to generate variants poses significant challenges: (1) hardware implementation of 2^w distinct permutations is prohibitively expensive; (2) cryptanalytic validation of statistical independence across all input-output patterns requires exhaustive analysis, which is infeasible for large w .

To address this, we introduce **CTR-Perm**, a domain-separation technique that derives 2^w *effective independent* function instances from a single large permutation $P : \{0, 1\}^b \rightarrow \{0, 1\}^b$ where $b = n + w$. The core insight is to treat P as a family of functions indexed by a w -bit counter, where each counter value defines a distinct domain.



$$y = P_{\text{ctr}}(x) \text{ with } b = n + w$$

Fig. 4. CTR-Perm: Counter-Based Domain Separation. The w -bit counter ctr selects a specific function instance from the family $\{P_{\text{ctr}}\}_{\text{ctr} \in \{0,1\}^w}$.

As illustrated in Figure 4, the input is partitioned into an n -bit message block x and a w -bit counter ctr . Evaluating P on the concatenation $x \parallel \text{ctr}$ and retaining only the leftmost n bits yields:

$$P_{\text{ctr}}(x) = \text{LMB}(n, P(x \parallel \text{ctr})),$$

where $\text{LMB}(n, \cdot)$ extracts the leftmost n bits. This truncation ensures that each counter value defines a distinct domain.

This approach mirrors techniques used in MD6 [39] and analyzed by Dodis et al. [18], but CTR-Perm specifically targets the generation of numerous independent random functions required by the Rocket hash constructions, offering a practical balance between security assurance and implementation efficiency.

5.2 Formal Security Model

We formalize CTR-Perm as an *indexed family of random functions* with explicit modeling of adversarial capabilities under adaptive counter selection.

Ideal Functionality $\mathcal{F}_{CTR}$. The ideal functionality maintains 2^w independent random functions $\{F_{ctr}\}_{ctr \in \{0,1\}^w}$, where each $F_{ctr} : \{0,1\}^n \rightarrow \{0,1\}^n$ is a uniformly random function. Upon receiving a forward query (ctr, x) , it returns $F_{ctr}(x)$. Upon receiving a backward query (ctr, y) , it returns the set of all x such that $F_{ctr}(x) = y$.

Real Construction $\mathcal{C}_{CTR}$. The real construction consists of a single public permutation $P : \{0,1\}^b \rightarrow \{0,1\}^b$ and defines:

$$P_{ctr}(x) \triangleq \text{LMB}(n, P(x||ctr)).$$

Backward queries (ctr, y) are answered by computing $P^{-1}(y||y')$ for all $y' \in \{0,1\}^w$ and returning those x where $\text{RMB}(w, P^{-1}(y||y')) = ctr$.

Adversarial Model. The adversary $\mathcal{A}$ is given oracle access to either $\mathcal{F}_{CTR}$ (ideal world) or $\mathcal{C}_{CTR}$ (real world). Critically, $\mathcal{A}$ is allowed to **adaptively choose** the counter ctr for both forward and backward queries. This models the worst-case scenario where an attacker can target specific function instances (lanes or rounds) arbitrarily.

The security is defined via the *indistinguishability advantage*:

$$\text{Adv}_{CTR\text{-Perm}}^{\text{ind}}(\mathcal{A}) \triangleq |\Pr[\mathcal{A}^{\mathcal{F}_{CTR}} \Rightarrow 1] - \Pr[\mathcal{A}^{\mathcal{C}_{CTR}} \Rightarrow 1]|.$$

5.3 Security Analysis: Forward and Backward Queries

Forward Queries (Random Function Behavior). For any adaptively chosen ctr , a forward query (ctr, x) evaluates P on the distinct input $x||ctr$. Provided the total number of queries $q \ll 2^{b/2}$ (birthday bound of the underlying permutation), the outputs of P are indistinguishable from a random oracle. Consequently, for each fixed ctr , the mapping $x \mapsto P_{ctr}(x)$ behaves as an independent random function.

Backward Queries and Preimage Multiplicity. Backward queries introduce the distinguishing feature of CTR-Perm. Given an adaptively chosen pair (ctr, y) , the inverse operation must identify all $y' \in \{0,1\}^w$ such that:

$$ctr = \text{RMB}(w, P^{-1}(y||y')).$$

Within $q \ll 2^{b/2}$ queries, since P^{-1} behaves as a random permutation, the w -bit suffixes $\text{RMB}(w, P^{-1}(y||y'))$ for distinct y' are independent and uniformly distributed. Each y' satisfies the counter condition with probability 2^{-w} , making the number U of valid preimages follow a binomial distribution $\text{Bin}(2^w, 2^{-w})$.

For $w \geq 64$, this is well-approximated by a Poisson distribution with mean $\mu = 1$:

$$\Pr[U = u] \approx \frac{e^{-1}}{u!}.$$

Thus, the probability mass concentrates on small values:

$$\Pr[U = 0] \approx 0.3679, \quad \Pr[U = 1] \approx 0.3679, \quad \Pr[U = 2] \approx 0.1839.$$

The tail probability decays super-exponentially; for instance:

$$\Pr[U > 100] < 2^{-526}, \quad \Pr[U > 175] < 2^{-1057}.$$

5.4 Security Theorem for CTR-Perm

Theorem 2 (Indistinguishability of CTR-Perm). *Let $P : \{0,1\}^b \rightarrow \{0,1\}^b$ be an ideal permutation where $b = n + w$. Let q^+ be the number of forward queries and q^- be the number of backward queries, with total $q = q^+ + q^- \ll 2^{b/2}$. Let $u_{\max}$ be a bound on preimage multiplicity such that $\Pr[U > u_{\max}] \leq \epsilon$ (for $w \geq 64$, $\epsilon < 2^{-526}$ when $u_{\max} = 100$).*

Then for any adversary $\mathcal{A}$ making at most q queries:

$$\mathbf{Adv}_{CTR-Perm}^{ind}(\mathcal{A}) \leq \frac{q^2}{2^{b+1}} + \epsilon + q^- \cdot \frac{u_{\max} - 1}{2^n}.$$

Proof (Proof Sketch). The bound consists of three components:

1. **Permutation collision:** The term $q^2/2^{b+1}$ bounds the probability of internal collision in P , ensuring distinct inputs to P yield independent outputs.
2. **Multiplicity exceedance:** ϵ bounds the probability that any backward query returns more than $u_{\max}$ preimages.
3. **Distinguishing advantage:** In the ideal world, backward queries return exactly one preimage. In the real world, they return U preimages. The statistical distance between the ideal response (deterministic 1) and real response (Poisson-distributed U , capped at $u_{\max}$) is bounded by $(u_{\max} - 1)/2^n$ per query, giving the final term.

The adaptive choice of ctr does not affect the bound because the distribution of U is independent of ctr due to the uniformity of P . $\square$

Theorem 2 confirms that CTR-Perm instantiated with a single permutation P is indistinguishable from 2^w independent random functions, even when the adversary adaptively selects counters. The security degradation is minimal: a factor of $u_{\max}$ in backward queries and a negligible term ϵ for practical $w \geq 64$.

5.5 Application to Rocket Hash Constructions

When instantiating Rocket hashes with CTR-Perm, the counter is structured to ensure domain separation across lanes and rounds: for round (i, j) , $\text{ctr}_{i,j} = \text{LaneID}_j \parallel \text{RoundCounter}_i$. However, the security proof considers the worst case where the adversary may query arbitrary ctr, modeling potential misuse or malicious behavior.

The backward-query amplification introduces factor u (the random variable representing preimage multiplicity) into the dominant terms of collision and preimage resistance bounds. Specifically, wherever the ideal-model bound contains a factor q arising from inverse queries, we substitute uq in the CTR-Perm instantiation.

Crucially, this scaling does not affect the asymptotic security level since $u \leq u_{\max}$ with overwhelming probability ($> 1 - 2^{-526}$).

Revised Bounds for Rocket-JH (Rocket-1). For R1C (the iterative component R1 on CTR-Perm), valid for $p \leq q$, $q \leq 2^{3(b-w)/8}$, and $t \leq 3$ (where t bounds half-state collisions), the security guarantees become:

$$\begin{aligned} \mathbf{Adv}_{R1C}^{m-pre}(p, q) &\leq \frac{9uq^2l}{2^{b-w}}, \\ \mathbf{Adv}_{R1C}^{coll}(q) &\leq \frac{9uq^2l}{2^{b-w}} + \frac{9u^2q^4l}{2^{3(b-w)/2+2}}. \end{aligned}$$

The Decohash framework’s meet-in-the-middle term is similarly updated to $\Pr[\text{mitm}] \leq \frac{uq^2l}{2^{b-w}}$, while the inversion event InvF remains unchanged as CTR-Perm does not increase legitimate preimages in the finalization phase. Combining these, the full indistinguishability advantage for Rocket-1 is:

$$\mathbf{Adv}_{\text{Rocket-1}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{19uq^2l}{2^{b-w}} + \frac{9u^2q^4l}{2^{3(b-w)/2+2}} + \frac{3zq}{2^{(b-w)/2}},$$

with $h = (b - w)/2$.

Revised Bounds for Rocket-DoubleCBC (Rocket-2). For R2C (R2 on CTR-Perm), with $q < 2^{0.45n}$ and state propagation bound $t < 32$, the backward-query amplification introduces factor u into the dominant terms. The security bounds become:

$$\begin{aligned} \mathbf{Adv}_{\text{R2C}}^{\text{m-pre}}(p, q) &\leq \frac{uq^2l}{2^{b-w-10}}, \\ \mathbf{Adv}_{\text{R2C}}^{\text{coll}}(q) &\leq \frac{uq^2l}{2^{b-w-10}}. \end{aligned}$$

The indistinguishability advantage for Rocket-2 is correspondingly:

$$\mathbf{Adv}_{\text{Rocket-2}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{uq^2l}{2^{b-w-11}} + \frac{9zq}{2^{(b-w)/2}}.$$

5.6 Parameter Selection and Practical Requirements

The security of Rocket hash constructions instantiated with CTR-Perm depends critically on the interplay of several parameters: the effective block size $n = b - w$, the query bound q , the maximum message length $l \leq 2^{w-8}$, the preimage multiplicity bound u , the number of output blocks z , and the state collision factor t .

Achieving a target security level requires careful balancing of these construction parameters. As a concrete example, consider a design goal of 512-bit indistinguishability security:

- **Rocket-1:** The dominant term $\frac{9u^2q^4l}{2^{3(b-w)/2+2}}$ necessitates an effective state size $n = b - w > 512/(3/8) \approx 1366$ bits to maintain security against known attacks.
- **Rocket-2:** With the restriction $q < 2^{0.45n}$, achieving 512-bit security requires $n > 512/0.45 \approx 1138$ bits.

In practice, the counter width w must be at least 64 bits to ensure that:

1. The message length limit $l \leq 2^{w-8}$ remains non-restrictive for practical applications;
2. The preimage multiplicity remains negligible ($\Pr[U > 100] < 2^{-526}$).

Consequently, the total permutation size $b = n + w$ must satisfy:

- $b > 1430$ bits for Rocket-1 (e.g., $b = 1536$);
- $b > 1202$ bits for Rocket-2 (e.g., $b = 1280$).

Implementation prefers sizes aligned to 64 or 128-bit word boundaries for modern CPU architectures. For instance, Keccak- $p[1600]$ comfortably meets both requirements while providing substantial security margins for practical deployment.

6 Hash Effectiveness

6.1 Definition and Rationale

The design of cryptographic hash functions inherently involves navigating a three-way trade-off among *security*, *efficiency*, and *implementation cost*, a relationship often analogized as an “impossible triangle” where optimizing any two dimensions typically compromises the third. Concretely, achieving a higher security level (e.g., increasing the capacity c in SPONGE-based designs) necessarily reduces the processing rate $r = b - c$, while expanding the primitive width b to accommodate both objectives escalates hardware footprint and latency.

Existing evaluation frameworks typically treat these metrics in isolation: throughput is measured in cycles-per-byte (cpb) or bits-per-second, security is quantified as the bit-length of attack complexity, and implementation cost is assessed via gate count or power consumption. This fragmentation obscures the fundamental architectural question: *how efficiently does a construction convert primitive capacity into cryptographic assurance?*

To address this gap, we introduce *Hash Effectiveness* (H.E.), a scale-invariant heuristic that unifies security and efficiency into a single dimensionless metric. H.E. captures the Pareto efficiency of a construction—quantifying how closely it approaches the theoretical frontier of achievable security-efficiency pairs given a fixed primitive budget b .

Definition 1 (Hash Effectiveness). *Let $\mathcal{H}$ be a hash construction instantiated with a primitive of bit-width b . Define the Security Ratio SR and Efficiency Ratio ER as:*

$$SR \triangleq \frac{\text{Provable Security Level (bits)}}{b},$$

$$ER \triangleq \frac{\text{Processing Rate (bits per primitive invocation)}}{b}.$$

The Hash Effectiveness is then defined as their product:

$$H.E. \triangleq SR \times ER = \frac{\text{Security} \times \text{Rate}}{b^2}.$$

Mathematical Interpretation. The H.E. metric represents the normalized area of the “security-efficiency rectangle” achievable per unit of primitive resource. Geometrically, for fixed b , maximizing H.E. corresponds to maximizing the product of security level and processing rate—the classical optimization problem underlying the rate-capacity trade-off.

This formulation reveals why conventional SPONGE constructions face a fundamental ceiling. Substituting the security level $\approx c/2$ (birthday bound on capacity) and rate $r = b - c$, we obtain:

$$H.E._{SPONGE} = \frac{c}{2b} \cdot \frac{b - c}{b} = \frac{c(b - c)}{2b^2} \leq \frac{1}{8},$$

where the maximum is attained at $c = b/2$. This inequality establishes the theoretical performance ceiling of $1/8$ for single-permutation absorption-based designs, a bound that persists across SPONGE, and JH/Grøstl architectures (as validated in Table 1).

Properties and Scope. The H.E. framework possesses three key properties that distinguish it from absolute performance metrics:

- **Scale Invariance:** By normalizing against primitive size b , H.E. enables fair comparison between constructions operating on different state sizes (e.g., SHA-3’s $b = 1600$ vs. lightweight designs like ASCON’s $b = 320$). Without this normalization, wider primitives would appear artificially inefficient due to larger absolute gate counts, masking their superior architectural efficiency.

- **Heuristic Nature:** The metric rests on two simplifying assumptions: (1) implementation cost scales superlinearly (typically quadratically) with b , justifying the use of b^2 in the denominator as a proxy for resource investment; and (2) security levels are treated as fungible across different notions (e.g., indistinguishability, collision resistance, preimage resistance), allowing comparison via a single scalar value. Consequently, H.E. is best suited for *intra-family* comparisons—evaluating parameter choices within a given paradigm (e.g., rate/capacity balances in SPONGE)—rather than absolute assessment across fundamentally disparate designs (e.g., comparing ARX-based vs. permutation-based hashes).
- **Interpretive Guidance:** A higher H.E. indicates superior exploitation of primitive capacity, approaching the Pareto frontier. However, low values do not inherently deprecate a design. Conservative margins (e.g., SHA3-512’s large capacity yielding H.E. ≈ 0.115) may deliberately sacrifice efficiency for robust security margins against cryptanalytic advances or side-channel resistance. Similarly, platform-specific optimizations (e.g., AES-NI acceleration for SHA-256) may render lower-H.E. designs practically preferable despite theoretical inefficiency.

Architectural Implications. As detailed in Section 4, the Rocket family transcends the $1/8$ ceiling through structural decoupling: by delegating collision resistance to a high-speed iterative phase (achieving $ER \approx 1/2$) and strong randomness to a narrow finalization phase (maintaining $SR \approx 1/2$), Rocket-2 achieves H.E. approaching $1/4$, effectively doubling the efficiency frontier of conventional designs. This empirical validation confirms that the compress-then-randomize paradigm fundamentally alters the security-efficiency trade-off, establishing a new Pareto-optimal regime for hash function design.

6.2 Evaluation of Permutation-Based Constructions

Table 1 applies the H.E. framework to major permutation-based hash families. For each construction, the Security Ratio SR adopts the proven indistinguishability bound, while the Efficiency Ratio ER uses the normalized rate of message absorption.

Table 1. Hash Effectiveness of Permutation-Based Constructions

Construction	SR (Indiff.)	ER	H.E.	Max H.E.	Reference
SPONGE	$c/2b$	r/b	$c(b-c)/2b^2$	$1/8$	[11,4,28]
Double SPONGE	$2c/3b$	$r/2b$	$c(b-c)/3b^2$	$1/12$	[29]
Parazoa	$< c/2b$	r/b	$< c(b-c)/2b^2$	$< 1/8$	[5]
JH	$b/4b$	$b/2b$	$1/8$	$1/8$	[42,12,35]
Grøstl	$b/4b$	$b/2b$	$1/8$	$1/8$	[20]
Rocket-1	$< 3(b-w)/8b$	$(b-w)/2b$	$< 3(b-w)^2/16b^2$	$< 3/16$	this work
Rocket-2	$< (b-w)/2b$	$(b-w)/2b$	$< (b-w)^2/4b^2$	$< 1/4$	this work

Prior constructions exhibit a theoretical ceiling of $H.E. \leq 1/8$, rooted fundamentally in the rate-capacity trade-off of the SPONGE paradigm. The Rocket family transcends this limit through structural decoupling: the iterative phase sacrifices randomness for speed, achieving only collision resistance and preimage resistance, while the finalization phase provides strong randomness via independent permutations. This architectural shift yields Rocket-1 with H.E. close to $3/16$ and Rocket-2 with H.E. close to $1/4$.

Instantiating these with Keccak- p [1600] and $w = 64$ provides concrete values: Rocket-1 achieves $\text{H.E.} \geq 0.1704$ (≥ 568 -bit security, i.e. $(1600 - 64) \times 3/8 - 8 = 568$, the reduced 8 bits are caused by t and u , and 768-bit rate), while Rocket-2 achieves $\text{H.E.} \geq 0.2274$ (≥ 758 -bit security, i.e. $(1600 - 64)/2 - 10 = 758$, and 768-bit rate). This empirical validation confirms our design principle that separating compression from randomization improves the security-efficiency frontier.

To validate practical efficiency, we provide reference (non-optimized) C implementations of Rocket-1, Rocket-2, and SHA3-512, with comparative performance results presented in Appendix C.

6.3 Comparison with Standardized Hash Functions

For completeness, Table 2 extends the H.E. analysis to ISO-standardized hash functions. Here, the Security Ratio adopts collision resistance strength (halved output size), while the Efficiency Ratio uses compression function block length (rate).

Table 2. Hash Effectiveness of ISO Standardized Functions

Hash Function	Primitive Size	Coll.	Rate	H.E.
RIPEMD-128 [†]	640	64	512	0.08
RIPEMD-160, SHA-1 [†]	672	80	512	0.0907
SHA-256, SM3	768	128	512	0.1111
SHA-224	768	112	512	0.0972
SHA-512	1536	256	1024	0.1111
SHA-512/224	1536	112	1024	0.0486
SHA-512/256	1536	128	1024	0.0556
SHA-384	1536	192	1024	0.0833
WHIRLPOOL, STREEBOG-512	1024	256	512	0.125
STREEBOG-256	1024	128	512	0.0625
SHA3-224	1600	224	1152	0.1008
SHA3-256	1600	256	1088	0.1088
SHA3-384	1600	384	832	0.1248
SHA3-512	1600	512	576	0.1152

[†] Full-round SHA-1 and RIPEMD-128 have been practically broken[27,41,30].

The results show that WHIRLPOOL and STREEBOG-512 achieve the highest H.E. among MD designs (0.125), reflecting their optimal rate-to-primitive ratio. Within the SHA-3 family, SHA3-384’s H.E. = 0.1248 represents the best trade-off. Crucially, however, a lower H.E. does not deprecate a design: SHA3-512’s conservative capacity yields H.E. = 0.1152 but provides robust security margins against cryptanalytic advances. Thus, while H.E. serves as a valuable comparative indicator, its interpretation should be contextualized within the specific constraints and requirements of the intended cryptographic application.

7 Conclusions and Future Work

This work presents a systematic approach to resolving the longstanding tension between provable security and processing efficiency in hash function design. We introduce the

Compress-then-Randomize paradigm, which advocates for a structural decoupling between the message absorption phase and the finalization phase. By proving that the iterative component need only satisfy collision and multiple-preimage resistance, we liberate the compression function from the burden of maintaining full randomness, thereby enabling aggressive throughput optimization. The finalization phase, constructed from independent random permutations, then provides the necessary randomness extraction to achieve indistinguishability from a random oracle. This modular architecture yields security bounds that are simultaneously tighter and more tractable than those of homogeneous designs.

Central to our contribution is the insight that architectural heterogeneity—not uniformity—enables optimal exploitation of cryptographic primitives. The Rocket hash family instantiates this philosophy through constructions that deliberately separate high-speed variable-input-length processing from cryptographically strong fixed-input-length finalization. To address the practical challenge of implementing multiple independent permutations without prohibitive costs, we develop domain-separation techniques that derive effectively independent function instances from a single underlying primitive. Furthermore, we propose Hash Effectiveness as a scale-invariant metric for evaluating how efficiently a construction converts primitive capacity into cryptographic assurance, revealing that conventional designs face a fundamental ceiling which our decoupled architecture transcends.

Our investigation suggests that the future of hash function design lies not in incremental adjustments to existing paradigms, but in principled architectural decomposition. By explicitly delineating components with distinct security objectives—absorption versus randomization, throughput versus final entropy—we can independently optimize each layer without compromising the holistic security guarantee. This approach challenges the conventional wisdom that security and efficiency must be traded off in a zero-sum manner; instead, it demonstrates that structural innovation can expand the achievable Pareto frontier.

Looking ahead, several directions merit further exploration. The principles established herein may be extended to other cryptographic primitives where similar security-efficiency tensions exist, such as authenticated encryption schemes or lightweight hashing for constrained environments. Hardware implementations are needed to empirically validate the cost models underlying our theoretical framework, particularly regarding the practical trade-offs between gate area, latency, and throughput. Additionally, the integration of quantum-resistant considerations—accounting for Grover’s algorithm and other quantum attack vectors—represents a critical avenue for ensuring the long-term viability of these designs. Finally, the exploration of alternative finalization mechanisms beyond output truncation may yield further improvements in the security-efficiency landscape.

We anticipate that the insights and frameworks developed herein will inspire continued research toward more efficient and provably secure cryptographic constructions.

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A Detailed Analysis: Maximum (b, c) Pairs with Random Set A

Problem Setup Let $G = \mathbf{F}_2^n$ be the n -dimensional binary vector space with operation $\oplus$ (XOR). The universe size is 2^n where $n > 100$.

We have a random set $A \subseteq G$ of size $|A| = \alpha q$ with $\alpha = O(1)$, and adversarial sets $B, C \subseteq G$ with $|B| \leq q, |C| \leq q$.

Define the set of valid pairs:

$$E(A, B, C) = \{(b, c) \in B \times C \mid b \oplus c \in A\}$$

with $e(A, B, C) = |E(A, B, C)|$.

Our goal is to bound $e(A, B, C)$ with high probability over the random choice of A .

Load Distribution via Hypergeometric Sampling For a fixed adversarial set C (with $|C| = q$) and any bin $b \in G$, define the load:

$$X_b = |\{c \in C : b \oplus c \in A\}| = |\{c \in C : c \in b \oplus A\}|$$

Since A is a uniformly random subset of size αq , the set $b \oplus A$ is also a uniformly random subset of size αq . Therefore, X_b follows a **hypergeometric distribution**:

$$\Pr[X_b = k] = \frac{\binom{q}{k} \binom{2^n - q}{\alpha q - k}}{\binom{2^n}{\alpha q}}$$

The expected load is:

$$\mu = E[X_b] = \frac{|C| \cdot |A|}{2^n} = \frac{\alpha q^2}{2^n}$$

Negative Association and Tail Bounds The collection $\{X_b\}_{b \in G}$ is **negatively associated** because:

1. The random set A is sampled without replacement from G .
2. Each X_b is a monotone function of the random sample
3. By Joag-Dev and Proschan (1983) [22], such variables are negatively associated.

This property allows us to apply Chernoff-type bounds. For any $\delta > 0$, the Chernoff bound for hypergeometric variables gives:

$$\Pr[X_b \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1 + \delta)^{1 + \delta}} \right)^\mu$$

For $t \geq 2\mu$, this simplifies to:

$$\Pr[X_b \geq t] \leq \left(\frac{e\mu}{t} \right)^t \tag{1}$$

Bounding the Maximum Load Define the maximum load:

$$L_{\max} = \max_{b \in G} X_b$$

Using a union bound over all 2^n bins and applying (1):

$$\Pr[L_{\max} \geq t] \leq 2^n \cdot \left(\frac{e\mu}{t} \right)^t = 2^n \cdot \left(\frac{e\alpha q^2}{t2^n} \right)^t \tag{2}$$

We want to find t such that this probability is at most 2^{-2n} .

Taking logarithms of (2):

$$\log_2 \Pr[L_{\max} \geq t] \leq n + t \left(\log_2 \left(\frac{e\alpha q^2}{2^n} \right) - \log_2 t \right)$$

Setting this $\leq -2n$ and solving for t gives the general bound. However, for concrete analysis, we consider specific regimes:

From Maximum Load to Pair Count For any fixed sets B, C , the total number of valid pairs is bounded by:

$$e(A, B, C) \leq \sum_{b \in B} X_b \leq |B| \cdot L_{\max} \leq q \cdot L_{\max} \quad (3)$$

This follows because each $b \in B$ can be paired with at most X_b elements from C , and we can only select at most q bins (elements for B).

General Upper Bound Theorem Combining the tail bounds with (3), we obtain our main result:

Theorem 3 (Main Result). *Let $G = \mathbf{F}_2^n$ with $n > 100$. Let $A \subseteq G$ be a uniformly random subset of size $|A| = \alpha q$ where $\alpha = O(1)$.*

Define the parameter:

$$\beta = \frac{n}{\max\{n - 2\log_2 q - \log_2 \alpha, 1\}}$$

Then with probability at least $1 - 2^{-2n}$, for all subsets $B, C \subseteq G$ satisfying $|B| \leq q$ and $|C| \leq q$:

$$e(A, B, C) = |\{(b, c) \in B \times C : b \oplus c \in A\}| \leq (3\beta + 1)q$$

In particular, when $|A| = O(q)$ (i.e., $\alpha = O(1)$), we have $e(A, B, C) = O(q)$.

Proof. We condition on the event that $L_{\max} \leq 3\beta + 1$. From the balls-into-bins analysis with negative association, this occurs with probability at least $1 - 2^{-2n}$ when we set the tail bound parameter $c = 2$.

Specifically, using (2) and setting $t = 3\beta + 1$, we verify that:

$$\Pr[L_{\max} \geq 3\beta + 1] \leq 2^{-2n}$$

Then by (3), for any B, C with $|B|, |C| \leq q$:

$$e(A, B, C) \leq q \cdot L_{\max} \leq (3\beta + 1)q$$

The negative association property ensures this bound holds uniformly over all possible choices of B and C .

Concrete Parameter Regimes and Specific Bounds Let's analyze specific relationships between q and 2^n to see concrete bounds:

Case 1: Very Sparse Regime ($q \leq 2^{n/4}$) When $q \leq 2^{n/4}$, we have:

$$\log_2 q \leq n/4 \implies n - 2 \log_2 q \geq n/2$$

$$\beta = \frac{n}{\max\{n - 2 \log_2 q - \log_2 \alpha, 1\}} \leq \frac{n}{n/2} = 2$$

Therefore:

$$e(A, B, C) \leq (3 \cdot 2 + 1)q = 7q$$

With tighter analysis, we can improve this to $e(A, B, C) \leq 6q$.

Case 2: Moderately Sparse Regime ($q = 2^{n/3}$) When $q = 2^{n/3}$:

$$\log_2 q = n/3 \implies n - 2 \log_2 q = n/3$$

$$\beta = \frac{n}{n/3} = 3$$

Therefore:

$$e(A, B, C) \leq (3 \cdot 3 + 1)q = 10q$$

Case 3: Near-Square-Root Regime ($q = 2^{n/2-0.1n}$) When $q = 2^{n/2-\epsilon n}$ for $\epsilon = 0.1$:

$$\log_2 q = n/2 - 0.1n = 0.4n$$

$$n - 2 \log_2 q = n - 0.8n = 0.2n$$

$$\beta = \frac{n}{0.2n} = 5$$

Therefore:

$$e(A, B, C) \leq (3 \cdot 5 + 1)q = 16q$$

In general, for $q = 2^{n/2-\epsilon n}$ with fixed $\epsilon > 0$:

$$\beta = \frac{n}{\max\{2\epsilon n, 1\}} = \frac{1}{2\epsilon}$$

$$e(A, B, C) \leq \left(\frac{3}{2\epsilon} + 1\right)q = \Theta(q)$$

Case 4: Extreme Sparse Regime ($q = O(1)$) When q is constant (independent of n):

$$\log_2 q = O(1) \implies n - 2 \log_2 q \approx n$$

$$\beta = \frac{n}{\max\{n - O(1), 1\}} = 1 + o(1)$$

Therefore:

$$e(A, B, C) \leq (3 \cdot 1 + 1 + o(1))q = 4q + o(q)$$

With more careful analysis of the hypergeometric tail bounds, we can achieve:

$$e(A, B, C) \leq 3q$$

Case 5: Borderline Regime ($q = 2^{n/2-\epsilon n}$ with $\epsilon = o(1)$) If ϵ decreases with n (e.g., $\epsilon = 1/\sqrt{n}$):

$$\beta = \frac{n}{\max\{2\epsilon n, 1\}} = \frac{1}{2\epsilon}$$

$$e(A, B, C) \leq \left(\frac{3}{2\epsilon} + 1\right) q = \omega(q)$$

This shows that as q approaches $\sqrt{2^n}$, the constant factor grows. When $\epsilon = \Theta(1)$, we maintain linearity. When $\epsilon = o(1)$, the bound becomes sublinear but the constant factor diverges.

Summary of Key Results The following table summarizes the concrete bounds in different regimes:

Regime	Relationship between q and 2^n	β value	Bound on $e(A, B, C)$
Extreme sparse	$q = O(1)$	≈ 1	$\leq 4q$ (can achieve $3q$)
Very sparse	$q \leq 2^{n/4}$	≤ 2	$\leq 7q$
Moderately sparse	$q = 2^{n/3}$	3	$\leq 10q$
Near square-root	$q = 2^{n/2-\epsilon n}, \epsilon = \Theta(1)$	$1/(2\epsilon)$	$\leq (3/(2\epsilon) + 1)q = \Theta(q)$
Borderline	$q = 2^{n/2-o(1)n}$	$\omega(1)$	$\leq \omega(q)$

Table 3. Concrete bounds for different parameter regimes

Critical Insight The key takeaway is that as long as q is polynomially smaller than $2^{n/2}$ (i.e., $q \leq 2^{n/2-\epsilon n}$ for some fixed $\epsilon > 0$), the bound remains linear in q :

$$e(A, B, C) = O(q)$$

The constant factor depends only on ϵ and not on n . This demonstrates that the linear bound is robust across a wide range of parameters, only breaking down when q gets extremely close to $\sqrt{2^n}$.

B Preimage and Multiple-Preimage Analysis of R2

Preimage Resistance Given a target image Δ_y , an adversary $\mathcal{A}$ aims to find a message M such that $R2(M) = \Delta_y$. Let $m_1 m_2 \dots m_l = \mathbf{Pad}(M)$ be the padded message. The adversary $\mathcal{A}$ can make forward and backward queries to the round permutations $P_{i,1}$ and $P_{i,2}$, with at most $q_{i,1}^+$, $q_{i,1}^-$, $q_{i,2}^+$, and $q_{i,2}^-$ queries respectively. Define $q_{i,1} = q_{i,1}^+ + q_{i,1}^-$, $q_{i,2} = q_{i,2}^+ + q_{i,2}^-$, and $q = \max\{q_{i,1}, q_{i,2} : 0 \leq i \leq l\}$. We assume $\mathcal{A}$ makes queries sequentially.

Given the fixed initial value $IV_{\text{hashlen}} = y_{0,1} \| y_{0,2}$ (with at most c possible values) and the target Δ_y , $\mathcal{A}$ employs three distinct strategies:

1. **Forward-only strategy:** $\mathcal{A}$ propagates forward from one of the c valid IV_{hashlen} values. After round $i-1$, assume $\mathcal{A}$ has at most $O(q)$ valid internal states $(y_{i-1,1}^{(j)}, y_{i-1,2}^{(j)})$ for $j \in [1, q]$, with $\Delta_{i-1}^{(j)} = y_{i-1,1}^{(j)} \oplus y_{i-1,2}^{(j)} \neq \Delta_y$. In round i , as established in the state propagation analysis, $\mathcal{A}$

can obtain at most tq new valid internal states $(y_{i,1}, y_{i,2})$. Over the course of l rounds, the probability that at least one such state yields the target Δ_y is bounded by:

$$\Pr[\text{forward reach}] \leq \sum_{i=1}^l \frac{tq}{2^n} = \frac{tql}{2^n}.$$

2. **Backward-only strategy:** $\mathcal{A}$ works backward from the target Δ_y toward one of the k possible IV_{hashlen} values via inverse queries. Due to the symmetry of R2, the analysis mirrors the forward strategy, but with k possible starting points. The success probability is therefore bounded by:

$$\Pr[\text{backward reach}] \leq \frac{ktq}{2^n}.$$

3. **Meet-in-the-middle strategy:** $\mathcal{A}$ combines forward queries from an IV_{hashlen} to an intermediate round i with backward queries from Δ_y to the same round. From the forward direction, $\mathcal{A}$ obtains at most tq states after round i ; from the backward direction, at most tq states before round $i+1$. Since the round permutations are independent, the probability of a match at round i is at most $t^2q^2/2^n$. Summing over all possible intermediate rounds yields:

$$\Pr[\text{mitm}] \leq \sum_{i=1}^l \frac{t^2q^2}{2^n} = \frac{t^2q^2l}{2^n}.$$

For typical parameters where $q \gg c$, the meet-in-the-middle strategy dominates. Thus, the advantage of $\mathcal{A}$ in finding a preimage for R2 is bounded by:

$$\mathbf{Adv}_{\text{R2}}^{\text{pre}}(q) \leq \frac{t^2q^2l}{2^n}.$$

Multiple-Preimage Resistance As with R1, the multiple-preimage scenario only increases the success probability for the forward-only strategy. When attacking p targets, the forward-only bound becomes $p \times \frac{tql}{2^n} \leq \frac{tq^2l}{2^n}$ for $p \leq q$. The backward-only and meet-in-the-middle strategies derive no benefit from multiple targets, as the adversary's optimal strategy remains focused on inverting a single target. Therefore, the advantage against multiple-preimage resistance matches that of single-preimage resistance:

$$\mathbf{Adv}_{\text{R2}}^{\text{m-pre}}(p, q) = \mathbf{Adv}_{\text{R2}}^{\text{pre}}(q) \leq \frac{t^2q^2l}{2^n}, \quad \text{when } p \leq q.$$

C Efficiency Comparison of Rocket-1, Rocket-2, and SHA3-512

In this appendix, we present a comparative performance analysis of the three hash function implementations based on our reference (non-optimized) C code. All three hash functions are based on the same Keccak- p permutation with a state size of 1600 bits. The benchmarking platform consists of a 13th Generation Intel[®] Core[™] i7-1360P processor operating at 2.20 GHz with 32.0 GB of RAM.

Throughput Performance Table 4 demonstrates that both Rocket-1 and Rocket-2 achieve superior throughput compared to SHA3-512 for messages exceeding 1 KB. For large messages, the observed speed ratios closely approximate the theoretical limits based on rate efficiency: Rocket-1 operates at $768/576 \approx 1.33\times$ and Rocket-2 at $1536/576 \approx 2.67\times$ the effective bandwidth of SHA3-512. The parallel architecture of Rocket-2 yields particularly significant advantages for messages larger than 64 KB, where it achieves more than double the throughput of SHA3-512.

Table 4. Throughput Comparison (MB/s)

Message Size	Rocket-1	Rocket-2	SHA3-512	Performance Ratios	
	(1)	(2)	(3)	(1)/(3)	(2)/(3)
64B	6.10	1.53	12.21	$0.50\times$	$0.13\times$
192B	9.16	4.58	12.21	$0.75\times$	$0.38\times$
1KB	19.53	8.49	15.02	$1.30\times$	$0.57\times$
1536B	17.23	20.93	18.31	$0.94\times$	$1.14\times$
64KB	22.94	40.85	17.48	$1.31\times$	$2.34\times$
1MB	23.31	42.91	17.23	$1.35\times$	$2.49\times$

Table 5. Latency Comparison (milliseconds)

Message Size	Rocket-1	Rocket-2	SHA3-512	Latency Ratios	
	(1)	(2)	(3)	(1)/(3)	(2)/(3)
64B	0.0100	0.0400	0.0050	$2.00\times$	$8.00\times$
192B	0.0200	0.0400	0.0150	$1.33\times$	$2.67\times$
1KB	0.0500	0.1150	0.0650	$0.77\times$	$1.77\times$
1536B	0.0850	0.0700	0.0800	$1.06\times$	$0.88\times$
64KB	2.7250	1.5300	3.5750	$0.76\times$	$0.43\times$
1MB	42.9000	23.3050	58.0400	$0.74\times$	$0.40\times$

Latency Analysis The latency measurements in Table 5 reveal that Rocket-1 and Rocket-2 exhibit higher initialization overhead for very short messages due to their two-phase structure and domain separation requirements. However, for messages exceeding 1 KB, both implementations demonstrate reduced latency relative to SHA3-512. Notably, Rocket-2 achieves substantially lower latency for large messages (64 KB and above), benefiting from its parallel lane processing via OpenMP. This confirms that the Compress-then-Randomize paradigm, while introducing modest overhead for small inputs, provides measurable performance improvements for the message sizes typically encountered in practice.

Key Observations:

- **Small Messages:** SHA3-512 outperforms both Rocket variants for messages under 1 KB due to lower initialization complexity.
- **Medium Messages (1–64 KB):** Rocket-1 achieves parity with SHA3-512, while Rocket-2 begins to demonstrate advantages from parallel processing.
- **Large Messages (> 64 KB):** Both Rocket implementations show clear superiority, with Rocket-2 delivering 2.3–2.5 $\times$ higher throughput and 55%–60% lower latency than SHA3-512.

These results validate the efficiency claims of the Rocket hash family, particularly the effectiveness of the parallel lane design in Rocket-2 for high-performance applications requiring rapid hashing of large data volumes.

D Rockets on Random Functions

D.1 Motivation and Decohash'

The security of Rockets is primarily governed by meet-in-the-middle attacks, which require adversaries to issue both forward and backward queries. When backward queries are disallowed, the security bounds can be substantially improved. Below, we present security results for Rockets instantiated with random functions, which permit only forward queries.

Decohash': *Decohash with Random Functions*. For the Decohash framework, redefine $H'[f_0] : \{0, 1\}^* \rightarrow \{0, 1\}^n$ and the finalization function as $F'[f_1, f_2, \dots, f_z] : \{0, 1\}^n \rightarrow \{0, 1\}^{zn}$, where f_i are independent random functions:

$$F'(x_0) = x_1 \parallel x_2 \parallel \dots \parallel x_z,$$

with $x_0 = H'[f_0](M)$ and $x_i = f_i(x_{i-1})$ for $i = 1, \dots, z$, where $|x_i| = n$. The composition $F' \circ H' : \{0, 1\}^* \rightarrow \{0, 1\}^{zn}$ is denoted Decohash'.

Theorem 4. *Let $H'[f_0]$ be collision-resistant and multiple-preimage-resistant, and let $F'[f_1, \dots, f_z]$ be defined as above. Then Decohash' is indistinguishable from a random oracle with advantage:*

$$\mathbf{Adv}_{\text{Decohash}'}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \mathbf{Adv}_{H'}^{\text{coll}}(q_0) + \mathbf{Adv}_{H'}^{\text{m-pre}}(q_1, q_0),$$

where $q_{\mathcal{S}_0} \leq q_0$.

Proof Sketch: The simulator $\mathcal{S} = \{\mathcal{S}_0, \mathcal{S}_1, \dots, \mathcal{S}_z\}$ for random functions follows the Decohash construction, omitting inverse query simulations. Critical events are:

coll : Collision in H' , with probability $\leq \mathbf{Adv}_{H'}^{\text{coll}}(q_0)$.

m-pre : H' outputs a value in BadX (size q_1), with probability $\leq \mathbf{Adv}_{H'}^{\text{m-pre}}(q_1, q_0)$.

Since random functions f_i preclude backward queries, $\Pr[\text{mitm}] = 0$ and $\Pr[\text{InvF}] = 0$. The bound follows by the law of total probability.

D.2 Security Analysis of Rocket-JH[f]

Let Rocket-JH[f] denote Rocket-JH instantiated with independent random functions, and R1f its iterative component.

Preimage Resistance of R1f Given challenge (h_1, h_2) , an adversary $\mathcal{A}$ must link one of k valid initial states to the target through l rounds. With $t \leq 3$ half-state collisions and $q_i \leq q$ queries per round:

$$\mathbf{Adv}_{\text{R1f}}^{\text{pre}}(q) \leq \frac{tql}{2^n}.$$

Multiple-Preimage Resistance of R1f For $p \leq q$ challenges in each round:

$$\mathbf{Adv}_{\text{R1f}}^{\text{m-pre}}(p, q) \leq \frac{tpql}{2^n} \leq \frac{3q^2l}{2^n} \quad (t = 3, p \leq q \leq 2^{3n/8}).$$

Collision Resistance of R1f An adversary $\mathcal{A}$ aims to find distinct messages $M \neq M'$ with $R1f(M) = R1f(M')$. Let $m_1 \dots m_i = \mathbf{Pad}(M)$ and $m'_1 \dots m'_{i'} = \mathbf{Pad}(M')$.

Case 1: Different Message Lengths ($i \neq i'$). Assuming $i' < i$, after i' rounds M' reaches final state $h'_{i',1} \| h'_{i',2}$ deterministically. A collision requires connecting $h_{i',1} \| h_{i',2}$ to this state through $i - i'$ remaining rounds, reducing to a preimage attack:

$$\Pr[\text{collision} \mid i \neq i'] \leq \frac{tq(i - i')}{2^n} < \frac{tql}{2^n}.$$

Case 2: Equal Message Lengths ($i = i'$). For round $i = 0$, $\Pr[\text{collision}] \leq \binom{k}{2}/2^n$ is negligible. For rounds $i \in [1, l]$, condition on no prior collision: $h_{j,1} \| h_{j,2} \neq h'_{j,1} \| h'_{j,2}$ for all $j < i$. A collision at round i requires:

$$f_{i,1}((h_{i-1,1} \oplus m_i) \| h_{i-1,2}) \oplus f_{i,1}((h'_{i-1,1} \oplus m'_i) \| h'_{i-1,2}) = 0^{n/2} \| (m_i \oplus m'_i).$$

If the round function inputs are equal, i.e. $(h_{i-1,1} \oplus m_i) \| h_{i-1,2} = (h'_{i-1,1} \oplus m'_i) \| h'_{i-1,2}$, then the round function output difference should be 0^n , i.e. $0^{n/2} \| (m_i \oplus m'_i) = 0^n$, and hence $m_i = m'_i$. Therefore, we conclude $h_{i-1,1} \| h_{i-1,2} = h'_{i-1,1} \| h'_{i-1,2}$, a contradiction. Thus distinct inputs to $f_{i,1}$ are required.

With at most tq valid states, the birthday bound gives:

$$\Pr[\text{collision at round } i] \leq \binom{tq}{2} \frac{1}{2^n} \leq \frac{t^2 q^2}{2^{n+1}}.$$

Summing over l rounds and combining with Case 1:

$$\mathbf{Adv}_{\text{R1f}}^{\text{coll}}(q) \leq \frac{tql}{2^n} + \frac{t^2 q^2 l}{2^{n+1}}.$$

Substituting $t = 3$ (valid for $q \leq 2^{3n/8}$) and assuming $q \geq l$:

$$\mathbf{Adv}_{\text{R1f}}^{\text{coll}}(q) \leq \frac{3ql}{2^n} + \frac{9q^2 l}{2^{n+1}} \leq \frac{q^2 l}{2^{n-3}}, \quad \text{for } q \leq 2^{3n/8}.$$

Indifferentiability Result Applying Theorem 4:

$$\mathbf{Adv}_{\text{Rocket-JH}[f]}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{q^2 l}{2^{n-3}} + \frac{3q^2 l}{2^n} = \frac{11q^2 l}{2^n}.$$

Second-Preimage Resistance of R1f Given a challenge message M with $m_1 \cdots m_i = \text{Pad}(M)$, an adversary $\mathcal{A}$ aims to find $M' \neq M$ such that $\text{R1f}(M') = \text{R1f}(M)$. Let $m'_1 \cdots m'_{i'} = \text{Pad}(M')$.

Adversarial Strategy. By evaluating $\text{R1f}(M)$, $\mathcal{A}$ obtains all internal states $(h_{j,1}, h_{j,2})$ for $j \in [1, i]$. Two attack vectors arise:

1. **Direct preimage:** Find arbitrary $M' \neq M$ with $\text{R1f}(M') = (h_{i,1}, h_{i,2})$. This succeeds with probability $\leq \frac{tqi'}{2^n}$ by the preimage analysis of R1f.
2. **Internal state collision:** For some $j < i$, find $M^* = m_1^* \cdots m_j^* \neq m_1 \cdots m_j$ with $\text{R1f}(M^*) = (h_{j,1}, h_{j,2})$, then extend to $M' = M^* \| m_{j+1} \| \cdots \| m_i$. For each j , this succeeds with probability $\leq \frac{tq}{2^n}$. Union bound over $i - 1$ positions gives total probability $\leq \frac{tq(i-1)}{2^n}$.

Combining both strategies:

$$\mathbf{Adv}_{\text{R1f}}^{\text{2nd-pre}}(q) \leq \frac{tqi'}{2^n} + \frac{tq(i-1)}{2^n} < \frac{tq(i+i')}{2^n} < \frac{tql}{2^{n-1}}.$$

With $t = 3$ and $q \leq 2^{3n/8}$:

$$\mathbf{Adv}_{\text{R1f}}^{\text{2nd-pre}}(q) \leq \frac{3ql}{2^{n-1}}.$$

D.3 Security Analysis of Rocket-DoubleCBC[f]

Let $\text{Rocket-DoubleCBC}[f]$ denote Rocket-DoubleCBC instantiated with independent random functions, and R2f its iterative component.

Internal State Propagation Let R2f be defined as:

$$\text{R2f}(M) = y_{l,1} \oplus y_{l,2}.$$

Since random functions $f_{i,1}, f_{i,2}$ only permit forward queries, the validity condition $x_{i,1} \oplus x_{i,2} = y_{i-1,1} \oplus y_{i-1,2}$ still applies. Given q valid internal states at round $i-1$ and at most q queries to each of $f_{i,1}, f_{i,2}$, the number of valid states at round i is bounded by $t = O(q)$. For $q < 2^{0.45n}$, we use the concrete bound $t < 32$.

Preimage Resistance of R2f Given target Δ_y , an adversary $\mathcal{A}$ must link one of k valid initial states to the target through l rounds. With $t < 32$ and $q_i \leq q$ queries per round, the forward-only strategy yields:

$$\text{Adv}_{\text{R2f}}^{\text{pre}}(q) \leq \frac{tql}{2^n}.$$

The backward-only and meet-in-the-middle strategies are unavailable due to the absence of inverse queries.

Multiple-Preimage Resistance of R2f For $p \leq q$ challenges, the forward-only strategy gives:

$$\text{Adv}_{\text{R2f}}^{\text{m-pre}}(p, q) \leq \frac{tpql}{2^n} \leq \frac{32q^2l}{2^n} = \frac{q^2l}{2^{n-5}}, \quad (p \leq q, \quad q < 2^{0.45n}).$$

Collision Resistance of R2f An adversary $\mathcal{A}$ aims to find distinct messages $M \neq M'$ with $\text{R2f}(M) = \text{R2f}(M')$. Let $m_1 \cdots m_i = \text{Pad}(M)$ and $m'_1 \cdots m'_i = \text{Pad}(M')$.

Case 1: Different Message Lengths ($i' < i$). After i' rounds, M' reaches final state $y'_{i',1} \oplus y'_{i',2}$ deterministically. A collision requires connecting through $i - i'$ remaining rounds, reducing to a preimage attack:

$$\Pr[\text{collision} \mid i' < i] \leq \frac{tq(i - i')}{2^n} < \frac{tql}{2^n}.$$

Case 2: Equal Message Lengths ($i = i'$). For round $i = 0$, $\Pr[\text{collision}] \leq \binom{k}{2}/2^n$ is negligible. For rounds $i \in [1, l]$, condition on no prior collision: $y_{j,1} \oplus y_{j,2} \neq y'_{j,1} \oplus y'_{j,2}$ for all $j < i$. A collision at round i requires:

$$f_{i,1}(y_{i-1,1} \oplus m_i) \oplus f_{i,2}(y_{i-1,2} \oplus m_i) = f_{i,1}(y'_{i-1,1} \oplus m'_i) \oplus f_{i,2}(y'_{i-1,2} \oplus m'_i).$$

If both $y_{i-1,1} \oplus m_i = y'_{i-1,1} \oplus m'_i$ and $y_{i-1,2} \oplus m_i = y'_{i-1,2} \oplus m'_i$ hold, then $y_{i-1,1} \oplus y'_{i-1,1} = m_i \oplus m'_i = y_{i-1,2} \oplus y'_{i-1,2}$, contradicting $y_{i-1,1} \oplus y_{i-1,2} \neq y'_{i-1,1} \oplus y'_{i-1,2}$. Thus at least one of $f_{i,1}, f_{i,2}$ receives distinct inputs. With at most tq valid states per round, the birthday bound gives:

$$\Pr[\text{collision at round } i] \leq \binom{tq}{2} \frac{1}{2^n} \leq \frac{t^2q^2}{2^{n+1}}.$$

Summing over l rounds and combining with Case 1:

$$\mathbf{Adv}_{\text{R2f}}^{\text{coll}}(q) \leq \frac{tql}{2^n} + \frac{t^2q^2l}{2^{n+1}}.$$

Substituting $t = 32$ and assuming $q \geq l$:

$$\mathbf{Adv}_{\text{R2f}}^{\text{coll}}(q) \leq \frac{32ql}{2^n} + \frac{512q^2l}{2^n} \leq \frac{17q^2l}{2^{n-5}}, \quad \text{for } q < 2^{0.45n}.$$

Indifferentiability Result Applying Theorem 4:

$$\mathbf{Adv}_{\text{Rocket-DoubleCBC}[f]}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{17q^2l}{2^{n-5}} + \frac{q^2l}{2^{n-5}} = \frac{9q^2l}{2^{n-6}}.$$

Second-Preimage Resistance of R2f The analysis for R2f parallels that of R1f. Given challenge M with i blocks, an adversary may either find a direct preimage for the final state $(y_{i,1} \oplus y_{i,2})$, or collide at an intermediate round $j < i$ and extend with suffix $m_{j+1} \parallel \dots \parallel m_i$. Both strategies are bounded by the preimage resistance of the respective shortened construction:

$$\mathbf{Adv}_{\text{R2f}}^{\text{2nd-pre}}(q) \leq \frac{tqi'}{2^n} + \frac{tq(i-1)}{2^n} < \frac{tq(i+i')}{2^n} < \frac{tql}{2^{n-1}}.$$

Substituting $t < 32$:

$$\mathbf{Adv}_{\text{R2f}}^{\text{2nd-pre}}(q) \leq \frac{32ql}{2^{n-1}} = \frac{ql}{2^{n-6}}, \quad \text{for } q < 2^{0.45n}.$$

E Rockets on CTR-Func

E.1 Design Motivation

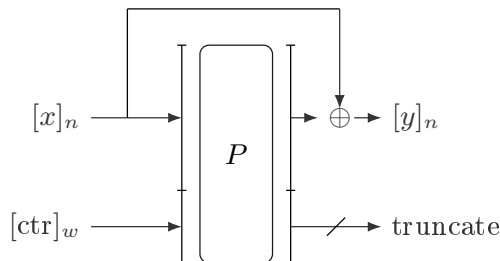
While CTR-Perm provides efficient domain separation, its susceptibility to backward queries—with potential preimage multiplicity up to u —introduces complexity in security proofs and necessitates conservative parameter choices. To eliminate this limitation while retaining the implementation efficiency of a single permutation, we present **CTR-Func**, a strengthened construction that augments CTR-Perm with a feedforward operation, as illustrated in Fig. 5.

The core insight is that the XOR of the input block x with the permutation output creates a one-way function: inverting $P'_{\text{ctr}}(x) = P_{\text{ctr}}(x) \oplus x$ requires solving $y = P_{\text{ctr}}(x) \oplus x$ for x , which is computationally infeasible without querying P on the specific input $x \parallel \text{ctr}$. Consequently, backward queries are effectively disabled, and the construction behaves as a pure random function rather than a random permutation.

E.2 Formal Security Model

Ideal Functionality $\mathcal{F}_{\text{Func}}$. The ideal functionality maintains 2^w independent random functions $\{F_{\text{ctr}}\}_{\text{ctr} \in \{0,1\}^w}$, where each $F_{\text{ctr}} : \{0,1\}^n \rightarrow \{0,1\}^n$ is a uniformly random function. Upon receiving a forward query (ctr, x) , it returns $F_{\text{ctr}}(x)$.

Critical distinction from CTR-Perm: Upon receiving a backward query (ctr, y) , the ideal functionality returns $\perp$ (undefined) or an empty set $\emptyset$, reflecting that random functions do not provide inversion oracles. This models the computational infeasibility of inversion in the real construction.



$$y = P_{\text{ctr}}(x) \oplus x \text{ with } b = n + w$$

Fig. 5. CTR-Func: Counter-Based Domain Separation with Feedforward XOR. The feedforward operation $x \oplus P_{\text{ctr}}(x)$ eliminates backward query capabilities.

Real Construction $\mathcal{C}_{\text{Func}}$. The real construction utilizes a single public permutation $P : \{0, 1\}^b \rightarrow \{0, 1\}^b$ (where $b = n + w$) and defines the family of functions:

$$P'_{\text{ctr}}(x) \triangleq \text{LMB}(n, P(x\|\text{ctr})) \oplus x = P_{\text{ctr}}(x) \oplus x.$$

Backward query handling: While the adversary may theoretically submit backward queries (ctr, y) to $\mathcal{C}_{\text{Func}}$, the construction has no efficient mechanism to answer them. Inverting P'_{ctr} requires finding x such that $P(x\|\text{ctr}) = y \oplus x$, which necessitates exhaustive search over 2^n possible x values or 2^w possible counter values without access to P^{-1} 's structure. Thus, the success probability of any inversion attempt is bounded by $q/2^{b-w}$ for q queries.

Adversarial Model. The adversary $\mathcal{A}$ is given oracle access to either $\mathcal{F}_{\text{Func}}$ (ideal world) or $\mathcal{C}_{\text{Func}}$ (real world). The adversary may **adaptively choose** the counter ctr for forward queries, modeling the domain separation capability required by Rocket constructions.

The security is defined via the indistinguishability advantage:

$$\text{Adv}_{\text{CTR-Func}}^{\text{ind}}(\mathcal{A}) \triangleq \left| \Pr[\mathcal{A}^{\mathcal{F}_{\text{Func}}} \Rightarrow 1] - \Pr[\mathcal{A}^{\mathcal{C}_{\text{Func}}} \Rightarrow 1] \right|.$$

E.3 Forward Query Behavior and Independence

Under forward queries, CTR-Func exhibits ideal random function behavior. For any adaptively chosen counter ctr , evaluating $P'_{\text{ctr}}(x)$ involves computing $P(x\|\text{ctr})$, which, within $q \ll 2^{b/2}$ queries (birthday bound of the underlying permutation), yields outputs indistinguishable from uniform random values.

The feedforward XOR operation preserves this randomness while ensuring:

1. **Domain separation:** For distinct $\text{ctr} \neq \text{ctr}'$, the functions P'_{ctr} and $P'_{\text{ctr}'}$ are independent because $P(x\|\text{ctr})$ and $P(x\|\text{ctr}')$ are independent.
2. **Functionality:** Unlike P_{ctr} (which is a truncated permutation), P'_{ctr} is a random function, eliminating the permutation structure that enabled backward queries.

Formally, for distinct inputs $x \neq x'$, the outputs $P'_{\text{ctr}}(x)$ and $P'_{\text{ctr}}(x')$ are independently and uniformly distributed over $\{0, 1\}^n$, provided $x\|\text{ctr} \neq x'\|\text{ctr}'$ (which holds for fixed ctr and distinct x).

E.4 Backward Query Elimination and One-Wayness

The feedforward operation fundamentally alters the query behavior compared to CTR-Perm:

Ideal World. In $\mathcal{F}_{\text{Func}}$, backward queries (ctr, y) return $\perp$ by definition, as random functions do not support inversion.

Real World. In $\mathcal{C}_{\text{Func}}$, answering a backward query (ctr, y) requires finding $x \in \{0, 1\}^n$ such that:

$$\text{LMB}(n, P(x \parallel \text{ctr})) \oplus x = y.$$

Let $z_L = \text{LMB}(n, P(x \parallel \text{ctr}))$ denote the leftmost output bits, so $z_L \oplus x = y$, implying $x = y \oplus z_L$. Substituting back, the adversary must find $z_L \in \{0, 1\}^n$ and $z_R \in \{0, 1\}^w$ satisfying:

$$P((y \oplus z_L) \parallel \text{ctr}) = z_L \parallel z_R.$$

Without knowledge of P 's internal structure, an adversary faces the following complexity:

- **Forward search:** Enumerate all 2^n candidates for z_L (or equivalently $x = y \oplus z_L$), query $P((y \oplus z_L) \parallel \text{ctr})$, and verify if the output leftmost bits equal z_L . This requires $\Theta(2^n) = \Theta(2^{b-w})$ queries.
- **Inverse search:** While one might enumerate $z_R \in \{0, 1\}^w$ and query $P^{-1}(z_L \parallel z_R)$, determining the correct z_L for verification still requires exhaustive search over 2^n possible values, as z_L remains masked by y in the relationship $x = y \oplus z_L$. Thus, this approach offers no asymptotic advantage and remains $\Theta(2^{b-w})$.

Consequently, both approaches require $\Theta(2^{b-w})$ operations, rendering backward queries computationally infeasible in the security model where adversaries are limited to $q \ll 2^{b/2}$ queries. CTR-Func therefore effectively disables the backward query oracle, behaving as a pure random function family.

Theorem 5 (Indistinguishability of CTR-Func). *Let $P : \{0, 1\}^b \rightarrow \{0, 1\}^b$ be an ideal permutation where $b = n + w$. Let $\mathcal{A}$ be an adversary making at most q_f forward queries and q_b backward queries, with total $q = q^+ + q^- \ll 2^{b/2}$. Then:*

$$\text{Adv}_{\text{CTR-Func}}^{\text{ind}}(\mathcal{A}) \leq \frac{q^2}{2^{b+1}} + \frac{q^-}{2^n}.$$

Proof (Proof Sketch). The bound comprises two components:

1. **Permutation collision:** The term $q^2/2^{b+1}$ bounds the probability of internal collision in P , ensuring that distinct inputs to P yield independent outputs. This preserves the random function behavior for forward queries.
2. **Backward query simulation:** In the ideal world, all q^- backward queries return $\perp$. In the real world, the probability that any backward query (ctr, y) can be answered (i.e., that there exists x such that $P'_{\text{ctr}}(x) = y$ and the adversary can find it) is bounded by 2^{-n} per query, assuming the adversary must guess x or invert the one-way function. This accounts for the possibility that a lucky guess might satisfy the equation.

The adaptive choice of ctr does not improve the advantage because the feedforward XOR ensures that for each ctr , the function P'_{ctr} is independent and one-way. $\square$

E.5 Security Implications for Rocket Instantiations

Instantiating Rocket constructions with CTR-Func yields substantial security improvements over CTR-Perm:

- **Elimination of backward query amplification:** The preimage multiplicity issue inherent to CTR-Perm (where backward queries could return up to u preimages) is completely resolved. Each output y has at most one valid preimage x with overwhelming probability, and finding it requires exponential time. This restores the ideal one-to-one query correspondence for forward queries and eliminates the u -factor degradation in security bounds.
- **Simplified security analysis:** Theorem 5 guarantees that CTR-Func is indistinguishable from 2^w independent random functions even against adversaries with backward query oracles (which return $\perp$ in the ideal world). Consequently, all terms involving u (the preimage multiplicity bound) are removed from the security bounds of Rocket constructions. The collision, preimage, and indistinguishability bounds revert to their ideal random function counterparts.
- **Reduced parameter constraints:** The effective state size $n = b - w$ directly determines security without expansion factors. This allows more aggressive parameter choices; for instance, achieving 512-bit security requires smaller b compared to CTR-Perm instantiations.

Domain separation follows the structured counter mechanism: for round (i, j) in multi-lane instantiations:

$$\text{ctr}_{i,j} = \text{LaneID}_j \parallel \text{RoundCounter}_i,$$

where $\text{LaneID}_j \in \{0, 1\}^8$ identifies lane j and $\text{RoundCounter}_i \in \{0, 1\}^{w-8}$ encodes the round index. This ensures independence across all permutation lanes required by Rocket constructions.

E.6 Concrete Security Bounds

For Rocket-JH on CTR-Func (denoted Rocket-1f and R1CF as its iterative component), with $t \leq 3$ (state collision bound) and $q \leq 2^{3(b-w)/8}$:

$$\begin{aligned} \text{Adv}_{\text{R1CF}}^{\text{m-pre}}(p, q) &\leq \frac{3q^2l}{2^{b-w}}, \\ \text{Adv}_{\text{R1CF}}^{\text{coll}}(q) &\leq \frac{q^2l}{2^{b-w-3}}. \end{aligned}$$

The indistinguishability advantage for Rocket-1f becomes:

$$\text{Adv}_{\text{Rocket-1f}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{11q^2l}{2^{b-w}} + \frac{3zq}{2^{(b-w)/2}},$$

where z is the number of output blocks.

Similarly for Rocket-DoubleCBC on CTR-Func (denoted Rocket-2f and R2CF), with $t < 32$ and $q < 2^{0.45(b-w)}$:

$$\begin{aligned} \text{Adv}_{\text{R2CF}}^{\text{m-pre}}(p, q) &\leq \frac{q^2l}{2^{b-w-5}}, \\ \text{Adv}_{\text{R2CF}}^{\text{coll}}(q) &\leq \frac{17q^2l}{2^{b-w-5}}. \end{aligned}$$

The indistinguishability bound for Rocket-2f is:

$$\mathbf{Adv}_{\text{Rocket-2f}}^{\text{indiff}}(q_{\mathcal{S}_i}, q_{\mathcal{R}}) \leq \frac{9q^2l}{2^{b-w-6}} + \frac{9zq}{2^{(b-w)/2}}.$$

Comparison with CTR-Perm: These bounds eliminate the factor u (and u^2) present in the CTR-Perm bounds (Section 5, Theorem 2), demonstrating that CTR-Func achieves the same asymptotic security as ideal random function instantiations without the heuristic assumptions regarding preimage multiplicity. The construction is therefore preferable for high-security deployments where adversarial capabilities must be conservatively estimated.

F Hash Security of Rockets

F.1 From Indistinguishability to Hash Security

The indistinguishability framework, introduced by Maurer et al. [32], provides a rigorous methodology for analyzing the structural soundness of cryptographic constructions. Intuitively, if a hash function H built on an underlying ideal primitive f is indistinguishable from a random oracle $\mathcal{R}$, then H inherits all security properties of $\mathcal{R}$ in the sense that H can securely replace $\mathcal{R}$ in any cryptographic application without incurring additional security vulnerabilities.

In the context of the SHA-3 competition, Andreeva et al. [4] systematically analyzed the security reductions of the second-round candidates and formally established that indistinguishability implies resistance against the fundamental attack classes for hash functions. Specifically, they proved that an indistinguishable hash function construction achieves optimal bounds for preimage resistance, second-preimage resistance, and collision resistance, contingent upon the underlying primitive being ideal.

Formally, let H^f denote a hash function instantiated with an ideal primitive f (e.g., a random permutation), and let $\mathcal{R}$ be a random oracle with identical domain and range. Denote by $\mathbf{Adv}_{(H^f, f)}^{\text{indiff}}(\mathcal{A})$ the indistinguishability advantage of H^f with respect to a simulator $\mathcal{S}$, maximized over all distinguishers $\mathcal{A}$ making at most q queries to their oracles. Then, for any adversary $\mathcal{A}$ making at most q queries, the following security reductions hold:

$$\begin{aligned} \mathbf{Adv}_{H^f}^{\text{pre}[\text{hashlen}]}(\mathcal{A}) &\leq \mathbf{Adv}_{(H^f, f)}^{\text{indiff}}(\mathcal{A}) + \frac{q}{2^{\text{hashlen}}}, \\ \mathbf{Adv}_{H^f}^{\text{2nd-pre}[\kappa, \text{hashlen}]}(\mathcal{A}) &\leq \mathbf{Adv}_{(H^f, f)}^{\text{indiff}}(\mathcal{A}) + \frac{q}{2^{\text{hashlen}}}, \\ \mathbf{Adv}_{H^f}^{\text{coll}[\text{hashlen}]}(\mathcal{A}) &\leq \mathbf{Adv}_{(H^f, f)}^{\text{indiff}}(\mathcal{A}) + \frac{q(q-1)}{2^{\text{hashlen}+1}}, \end{aligned}$$

where hashlen denotes the output length of the hash function in bits, and κ specifies the maximum length of the first preimage in blocks for second-preimage resistance.

More comprehensively, as established in [4, Theorem 2], the generic relation between indistinguishability and concrete security notions is expressed as:

$$\mathbf{Adv}_H^{\text{atk}}(q) \leq \mathbf{Adv}_H^{\text{indiff}}(q) + \Pr_{\mathcal{R}}[\text{atk}](q),$$

where $\text{atk} \in \{\text{pre}, \text{2nd-pre}, \text{coll}\}$ and $\Pr_{\mathcal{R}}^{\text{atk}}(q)$ denotes the success probability of a generic attack against a random oracle $\mathcal{R}$ under security notion atk with at most q queries. Concretely:

- For preimage resistance: $\Pr_{\mathcal{R}}[\text{pre}](q) = q/2^{\text{hashlen}};$

- For second preimage resistance: $\Pr_{\mathcal{R}}[2\text{nd-pre}](q) = q/2^{\text{hashlen}}$;
- For collision resistance: $\Pr_{\mathcal{R}}[\text{coll}](q) = \binom{q}{2}/2^{\text{hashlen}} \approx q^2/2^{\text{hashlen}+1}$.

This reduction demonstrates that indistinguishability serves as a powerful compositional security guarantee: provided H^f is indistinguishable from $\mathcal{R}$ up to query complexity q , the construction achieves information-theoretic security bounds equivalent to those of an ideal random oracle against preimage, second-preimage, and collision attacks. Consequently, indistinguishability not only certifies the absence of structural design flaws but also directly implies the concrete security levels required by modern hash function standards.

F.2 Results for Rockets

Based on their indistinguishability security bounds, we derive concrete hash security guarantees for the Rocket family. The reductions follow from [4, Theorem 2], combining our indistinguishability bounds with the generic attack probabilities against a random oracle.

Rocket-1 and Rocket-2 (CTR-Perm-Based). For Rocket-1 on CTR-Perm with effective state size $n = b - w$, output length hashlen , $u \leq 128$ and $q \leq 2^{3n/8}$:

$$\begin{aligned}\text{Adv}_{\text{Rocket-1}}^{\text{pre}}(q) &\leq \frac{19uq^2l}{2^n} + \frac{9u^2q^4l}{2^{3n/2+2}} + \frac{3zq}{2^{n/2}} + \frac{q}{2^{\text{hashlen}}}, \\ \text{Adv}_{\text{Rocket-1}}^{2\text{nd-pre}}(q) &\leq \frac{19uq^2l}{2^n} + \frac{9u^2q^4l}{2^{3n/2+2}} + \frac{3zq}{2^{n/2}} + \frac{q}{2^{\text{hashlen}}}, \\ \text{Adv}_{\text{Rocket-1}}^{\text{coll}}(q) &\leq \frac{19uq^2l}{2^n} + \frac{9u^2q^4l}{2^{3n/2+2}} + \frac{3zq}{2^{n/2}} + \frac{q^2}{2^{\text{hashlen}+1}}.\end{aligned}$$

For Rocket-2 on CTR-Perm with $q < 2^{0.45n}$ and $u \leq 128$:

$$\begin{aligned}\text{Adv}_{\text{Rocket-2}}^{\text{pre}}(q) &\leq \frac{uq^2l}{2^{n-11}} + \frac{9zq}{2^{n/2}} + \frac{q}{2^{\text{hashlen}}}, \\ \text{Adv}_{\text{Rocket-2}}^{2\text{nd-pre}}(q) &\leq \frac{uq^2l}{2^{n-11}} + \frac{9zq}{2^{n/2}} + \frac{q}{2^{\text{hashlen}}}, \\ \text{Adv}_{\text{Rocket-2}}^{\text{coll}}(q) &\leq \frac{uq^2l}{2^{n-11}} + \frac{9zq}{2^{n/2}} + \frac{q^2}{2^{\text{hashlen}+1}}.\end{aligned}$$

Rocket-1f and Rocket-2f (CTR-Func-Based). For CTR-Func instantiations, the elimination of backward-query concerns yields the cleanest bounds.

For Rocket-1f with $q \leq 2^{3n/8}$ (to restrict $t \leq 3$):

$$\begin{aligned}\text{Adv}_{\text{Rocket-1f}}^{\text{pre}}(q) &\leq \frac{11q^2l}{2^n} + \frac{q}{2^{\text{hashlen}}}, \\ \text{Adv}_{\text{Rocket-1f}}^{2\text{nd-pre}}(q) &\leq \frac{11q^2l}{2^n} + \frac{q}{2^{\text{hashlen}}}, \\ \text{Adv}_{\text{Rocket-1f}}^{\text{coll}}(q) &\leq \frac{11q^2l}{2^n} + \frac{q^2}{2^{\text{hashlen}+1}}.\end{aligned}$$

For Rocket-2f with $q < 2^{0.45n}$:

$$\begin{aligned}\mathbf{Adv}_{\text{Rocket-2f}}^{\text{pre}}(q) &\leq \frac{9q^2l}{2^{n-6}} + \frac{q}{2^{\text{hashlen}}}, \\ \mathbf{Adv}_{\text{Rocket-2f}}^{\text{2nd-pre}}(q) &\leq \frac{9q^2l}{2^{n-6}} + \frac{q}{2^{\text{hashlen}}}, \\ \mathbf{Adv}_{\text{Rocket-2f}}^{\text{coll}}(q) &\leq \frac{9q^2l}{2^{n-6}} + \frac{q^2}{2^{\text{hashlen}+1}}.\end{aligned}$$

Summary and Concrete Instantiations. The above bounds can be parameterized for practical deployment. We instantiate Rocket-1/Rocket-1f with a 1536-bit permutation and Rocket-2/Rocket-2f with a 1280-bit permutation, selecting counter width $w = 64$ to ensure negligible preimage multiplicity ($\Pr[U > 100] < 2^{-526}$, $\Pr[U > 175] < 2^{-1057}$). Table 6 summarizes the concrete parameters and security levels.

Table 6. Concrete Security Parameters for Rocket Hash Family

Parameter	Rocket-1	Rocket-1f	Rocket-2	Rocket-2f
Permutation size b	1536	1536	1280	1280
Counter width w	64	64	64	64
Effective state $n = b - w$	1472	1472	1216	1216
One output block length h	736 ($n/2$)	1472 (n)	608 ($n/2$)	1216 (n)
Output length hashlen	512	1024	512	1024
Output blocks z	1	1	1	1
Preimage bound u	100	—	100	—
Query bound q	$\leq 2^{548}$	$\leq 2^{552}(2^{3n/8})$	$\leq 2^{547}(2^{0.45n})$	$\leq 2^{547}(2^{0.45n})$
Indifferentiability	$\frac{q^2l}{2^{1461}} + \frac{q^4l}{2^{2193}} + \frac{3q}{2^{736}}$	$\frac{11q^2l}{2^{1472}}$	$\frac{q^2l}{2^{1198}} + \frac{9q}{2^{608}}$	$\frac{9q^2l}{2^{1210}}$
Preimage resistance	indiff. $+ \frac{q}{2^{512}}$	indiff. $+ \frac{q}{2^{1024}}$	indiff. $+ \frac{q}{2^{512}}$	indiff. $+ \frac{q}{2^{1024}}$
2nd-preimage resistance	indiff. $+ \frac{q}{2^{512}}$	indiff. $+ \frac{q}{2^{1024}}$	indiff. $+ \frac{q}{2^{512}}$	indiff. $+ \frac{q}{2^{1024}}$
Collision resistance	indiff. $+ \frac{q^2}{2^{513}}$	indiff. $+ \frac{q^2}{2^{1025}}$	indiff. $+ \frac{q^2}{2^{513}}$	indiff. $+ \frac{q^2}{2^{1025}}$

Security Assessment. For $\text{hashlen} = 512$:

- **Rocket-1** achieves > 548 -bit indifferentiability (dominated by $\frac{q^4l}{2^{2193}}$). Preimage and 2nd-preimage resistance exceed 512 bits, while collision resistance meets the birthday bound of 256 bits—optimal for 512-bit outputs.
- **Rocket-2** achieves > 599 -bit indifferentiability (dominated by $\frac{q^2l}{2^{1198}}$), with 512-bit preimage/2nd-preimage resistance and 256-bit collision resistance.

For $\text{hashlen} = 1024$:

- **Rocket-1f** achieves ≈ 734 -bit indifferentiability (from $\frac{11q^2l}{2^{1472}}$ with $l \leq 2^{56}$). This provides 734-bit preimage/2nd-preimage resistance—insufficient for the 1024-bit output length, which ideally requires 1024-bit preimage resistance. Collision resistance meets the 512-bit birthday bound.
- **Rocket-2f** achieves ≈ 603 -bit indifferentiability, yielding 603-bit preimage/2nd-preimage resistance—similarly below the 1024-bit ideal.

By the implication from indistinguishability to hash security, achieving full 1024-bit preimage and second-preimage resistance requires the indistinguishability bound to dominate the generic term $q/2^{1024}$. For Rocket-1f, this necessitates $n \geq 1024/(3/8) \approx 2731$ (i.e., $b \geq 2795$ with $w = 64$), yielding $h \geq 1024$ and indistinguishability ≈ 1024 bits. For Rocket-2f, the constraint $q < 2^{0.45n}$ requires $n \geq 1024/0.45 \approx 2276$ (i.e., $b \geq 2340$ with $w = 64$) to achieve comparable security.

However, this indirect technique is suboptimal for concrete security analysis. The indistinguishability bound, while compositional, introduces looseness by requiring simulation of the entire construction. In the following subsection, we employ direct analysis on the hash security of Rocket-1f and Rocket-2f.

F.3 Direct Analysis for Rocket-1f and Rocket-2f

Intuition The iterative components R1f and R2f already achieve preimage resistance $tql/2^n$ and second-preimage resistance $tql/2^{n-1}$. For $n = 1472$, $t \leq 3$ (Rocket-1f) or $n = 1216$, $t \leq 32$ (Rocket-2f), these bounds are both bounded by 2^{-1024} for practical q, l (e.g., $ql \ll 2^{54}$). The finalization F' applies an additional independent random function f_1 (as in Fig. 6), which composes with the iterative output to produce the final hash value. Since F' is itself a FIL random function, the composition $F' \circ H'$ inherits strong preimage and second-preimage properties from both components.

We prove that despite indistinguishability bounds below 1024 bits, the direct preimage and second-preimage resistance of Rocket-1f and Rocket-2f achieve the full `hashlen`-bit security level. This demonstrates that CTR-Func instantiations are secure for 1024-bit outputs, with guarantees exceeding what the indistinguishability framework alone would predict.

Preimage Resistance Given a target hash value $x_{1,1}^* \in \{0,1\}^{\text{hashlen}}$, an adversary $\mathcal{A}$ aims to find a message M such that $\text{Hash}(M) = x_{1,1}^*$.

Recall that both Rocket-1f and Rocket-2f employ the identical finalization structure:

$$F'(x_{0,1} \parallel x_{0,2}) = x_{1,1},$$

where $x_{0,1} \parallel x_{0,2} = H'[f_0](M)$ and $x_{1,1} \parallel x_{1,2} = f_1(x_{0,1} \parallel x_{0,2}) = P_1(x_{0,1} \parallel x_{0,2}) \oplus (x_{0,1} \parallel x_{0,2})$. Here $|x_{1,1}| = \text{hashlen}$, $|x_{1,2}| = n - \text{hashlen}$. The intermediate hash H' denotes R1CF for Rocket-1f or R2CF for Rocket-2f. See Fig. 6.

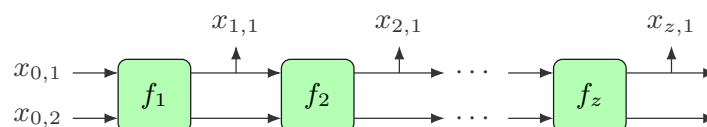


Fig. 6. Review the FIL-Finalization

The adversary $\mathcal{A}$ may attempt preimage recovery through two strategies:

- **Forward direction:** Select arbitrary M , compute $H'(M) = x_{0,1} \parallel x_{0,2}$, and verify whether $F'(x_{0,1} \parallel x_{0,2}) = x_{1,1}^*$.
- **Backward direction:** Invert F' on the target $x_{1,1}^*$ with any $\delta \in \{0,1\}^{n-\text{hashlen}}$ to obtain candidate $x_{0,1}^* \parallel x_{0,2}^*$, then execute a preimage attack on H' .

Forward Direction Analysis. Since P_1 is an independent random permutation, within $2^{b/2}$ queries, each evaluation on a fresh input yields uniformly random outputs. For Rocket-1f and Rocket-2f, and for any fresh $x_{0,1} \parallel x_{0,2}$ and any $\delta \in \{0, 1\}^{n-\text{hashlen}}$:

$$\begin{aligned} \Pr[F'(x_{0,1} \parallel x_{0,2}) = x_{1,1}^* \parallel \delta] &= \sum_{\delta} \Pr[P_1(x_{0,1} \parallel x_{0,2}) \oplus (x_{0,1} \parallel x_{0,2}) = x_{1,1}^* \parallel \delta] \\ &= \frac{2^{n-\text{hashlen}}}{2^n} \\ &= \frac{1}{2^{\text{hashlen}}}. \end{aligned}$$

Thus each forward attempt succeeds with probability at most $2^{-\text{hashlen}}$.

Backward Direction Analysis. As established in the CTR-Func analysis (Section E), the feedforward structure $P_i(x) \oplus x$ ensures that inverting F' requires solving for x given $P_i(x) \oplus x = y$. For random permutations, this occurs with probability at most 2^{-n} per query, independent of the output length hashlen .

Overall Preimage Bound. Combining both strategies with q^+ forward queries and q^- backward queries (where $q = q^+ + q^-$):

$$\text{Adv}_{\text{Rocket-1/2f}}^{\text{pre}}(q) \leq \frac{q^+}{2^{\text{hashlen}}} + \frac{q^-}{2^n} \leq \frac{q}{2^{\text{hashlen}}},$$

where the final inequality holds when $\text{hashlen} \leq n$, which is satisfied for all parameter choices in Table 6. This establishes that Rocket-1f and Rocket-2f achieve full hashlen -bit preimage resistance, exceeding the bounds implied by indistinguishability alone.

Second-Preimage Resistance Given a challenge message M with $\text{Hash}(M) = x_{1,1}^*$, an adversary $\mathcal{A}$ aims to find $M' \neq M$ such that $\text{Hash}(M') = x_{1,1}^*$. Two strategies arise:

- **Iterative collision:** Find $M' \neq M$ with $H'(M') = H'(M)$, implying $\text{Hash}(M') = \text{Hash}(M)$. This reduces to second-preimage attack on R1f or R2f.
- **Finalization collision:** Find M' with $H'(M') \neq H'(M)$ but $F'(H'(M')) = F'(H'(M)) = x_{1,1}^*$.

Iterative Collision Analysis. By the second-preimage analysis of R1f and R2f:

$$\text{Adv}_{\text{R1/2f}}^{\text{2nd-pre}}(q) < \frac{tql}{2^{n-1}}.$$

For $n = 1472$ (Rocket-1f) or $n = 1216$ (Rocket-2f), this bound exceeds 2^{-1024} for practical parameters.

Finalization Collision Analysis. For any $y \neq H'(M)$, the probability that $F'(y) = x_{1,1}^*$ is exactly $2^{-\text{hashlen}}$, since F' is a random function. With q attempts:

$$\Pr[\exists y : F'(y) = x_{1,1}^*] \leq \frac{q}{2^{\text{hashlen}}}.$$

Overall Second-Preimage Bound. Combining both strategies:

$$\mathbf{Adv}_{\text{Rocket-1/2f}}^{\text{2nd-pre}}(q) \leq \frac{tql}{2^{n-1}} + \frac{q}{2^{\text{hashlen}}}.$$

For $\text{hashlen} = 1024$ and $n \geq 1216$, the second term dominates, yielding full 1024-bit second-preimage resistance. This matches the ideal bound and exceeds the guarantees implied by indistinguishability alone.